UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA

Department of Information Technology Studies • IT Service Management

CHAPTER 5 (PART A)

# ITIL Management Practices

The 34 practices of the ITIL SVS, and the fourteen general management practices in detail — from architecture management to workforce and talent management.

Lecture slides adapted for UPSA from ITIL® Foundation, ITIL 4 Edition (AXELOS) — introduction and sections 5.1 to 5.1.14.1

SESSION ROADMAP

# What we cover in this session

1

## What a practice is

Definition, the three origins, and the full list of 34 practices.

2

## High-velocity service delivery

Why speed to market is reshaping every practice.

3

## Direction-setting practices

Architecture, strategy, portfolio and project management.

4

## Improvement and knowledge

Continual improvement, knowledge management, measurement and reporting.

5

## Protection practices

Information security management and risk management.

6

## Relationship practices

Relationship management and supplier management, including service integration.

7

## Money and people

Service financial management, organisational change management, workforce and talent.

8

## Contemporary issues

AI, FinOps, cyber resilience, multi-sourcing and the Ghanaian context.

# By the end of this session you should be able to…

1

## Define a practice

State what a management practice is and explain how practices relate to the service value chain.

2

## Classify the practices

Distinguish general, service and technical management practices and explain where each group came from.

3

## Recall the count

Identify the 14 general, 17 service and 3 technical management practices that make up the 34.

4

## Explain high velocity

Describe the high-velocity service delivery paradigm and its implications for practices.

5

## State the purposes

Give the purpose of each of the fourteen general management practices in your own words.

6

## Read a heat map

Interpret the contribution of a practice to each of the six value chain activities.

# By the end of this session you should also be able to…

7

## Apply architecture thinking

Identify the five architecture domains and what each contributes to organisational agility.

8

## Use improvement tooling

Explain the continual improvement register and the techniques used to assess a current state.

9

## Balance security

Explain prevention, detection and correction, and the trade-off with innovation.

10

## Analyse sourcing

Compare insourcing, outsourcing, single sourcing and multi-sourcing, and explain service integration.

11

## Critique metrics

Explain why individual KPIs can drive the wrong behaviour and what to do about it.

12

## Evaluate current issues

Assess how AI, cloud economics and regulation are reshaping the general management practices.

WHERE THIS FITS

# From Chapter 4 to Chapter 5

- Chapter 4 established the service value system: guiding principles, governance, the service value chain, practices and continual improvement.
- The value chain activities convert inputs into outputs — but they need resources to do the work. Those resources are the practices.
- Every value chain activity draws on a different combination of practices, and the same practice serves many activities.
- This is why ITIL 4 abandoned the language of processes owned by departments: a practice is broader than a process, and is shared.
- Chapter 5 works through all 34; this session covers the introduction and the fourteen general management practices.

DEFINITION TO MEMORISE

A management practice is a set of organisational resources designed for performing work or accomplishing an objective.

# Three families of practices, three different origins

*All 34 practices are subject to the four dimensions of service management introduced in Chapter 3.*

## General management practices

Adopted and adapted for service management from general business management domains — strategy, finance, projects, risk, people.

Fourteen practices.

## Service management practices

Developed within the service management and ITSM industries themselves — incident, change control, service desk, service level and the rest.

Seventeen practices.

## Technical management practices

Adapted from technology management domains by expanding or shifting their focus from technology solutions to IT services.

Three practices.

TABLE 5.1

# The 34 ITIL management practices

## General (14)

- Architecture management
- Continual improvement
- Information security management
- Knowledge management
- Measurement and reporting
- Organizational change management
- Portfolio management
- Project management
- Relationship management
- Risk management
- Service financial management
- Strategy management
- Supplier management
- Workforce and talent management

## Service (17)

- Availability management
- Business analysis
- Capacity and performance management
- Change control
- Incident management
- IT asset management
- Monitoring and event management
- Problem management
- Release management
- Service catalogue management
- Service configuration management
- Service continuity management
- Service design
- Service desk
- Service level management
- Service request management
- Service validation and testing

## Technical (3)

- Deployment management
- Infrastructure and platform management
- Software development and management

# The anatomy of a practice — and of a heat map

## Every practice is presented the same way

- A key message stating its purpose
- Scope: what the practice covers and what it does not
- Core activities and, in some cases, definitions
- Its contribution to each of the six value chain activities

## The heat map

Each practice is illustrated with a diagram of the value chain shaded by contribution. The scale runs from 0 (no contribution) through 1 (low) and 2, to 3 (high). It answers one question: where does this practice do most of its work?

## Why the heat map matters

Two practices may both 'contribute to all activities' yet behave completely differently. Architecture management concentrates in plan, improve and design and transition; knowledge management concentrates in deliver and support.

## A caution for revision

The heat maps are indicative, not examinable numbers. What matters is understanding why a practice is heavy in one activity and light in another.



ITSM IN THE MODERN WORLD

# High-velocity service delivery

Speed to market has become a condition of survival — and it changes every practice in the SVS.

# The paradigm in seven elements

1

## Fast delivery

New and changed IT services reach users quickly.

2

## Continual feedback analysis

Feedback on services is analysed at every stage of their lifecycle.

3

## Agility in processing it

Feedback is acted on quickly, giving continual and fast service improvement.

4

## End-to-end lifecycle view

From ideation, through creation and delivery, to consumption of services.

5

## Integrated product and service management

Product and service management practices are brought together rather than run in parallel.

6

## Digitalised infrastructure

IT infrastructure is digitalised and cloud computing is adopted.

7

## Extensive automation

The service delivery chain is automated as far as is sensible.

## Why it arose

If an organisation takes too long to implement a new business idea, someone else will implement it faster. Business has therefore begun demanding shorter time to market from IT.

# Who finds this easy, and who does not

## Providers born modern

- Have always used modern technology and modern ways of scaling resources.
- Already hold practices for project and product management, testing, integration, deployment, release, delivery and support.
- Documenting those practices helped trigger movements such as DevOps.
- For them, shorter time to market has not been a major challenge.

## Providers carrying legacy

- Bear a legacy of older IT architectures and management practices built for control and cost efficiency.
- The new business demand presents a far greater challenge.
- Significant organisational change is required even if only part of the portfolio needs high velocity.
- Bi-modal IT — high-velocity management alongside traditional practices — adds complexity rather than removing it.

# What an organisation must then consider

### Agile project management

Iterative delivery rather than long fixed plans.

### Agile financial management

Funding that flexes with consumption instead of annual fixed forecasts.

### Product-based structure

Teams organised around products rather than technical silos.

### Adaptive risk, audit and compliance

Controls that keep pace with rapid change rather than blocking it.

### Flexible architecture management

Architecture that enables change instead of freezing it.

### Specific technology choices

Architectural solutions such as microservices.

### Complex partner environments

Many suppliers, integrated deliberately rather than accidentally.

### Continual technology monitoring

Watching innovations and experimenting with them.

### Human-centred design

Designing around the people who will actually use the service.

### Cloud-focused infrastructure

Infrastructure management oriented to cloud computing.

### It touches every family

High-velocity delivery influences general, service and technical management practices alike.

### For discussion

Which two of these would be hardest to introduce in an organisation you know, and why?

A

SECTION 5.1

# The fourteen general management practices

Adopted and adapted for service management from general business management domains.

5.1

# The fourteen at a glance

### Architecture management

Understand the elements of the organisation and how they interrelate.

### Continual improvement

Align practices and services with changing business needs.

### Information security management

Protect the information the organisation needs to do business.

### Knowledge management

Improve effective, efficient and convenient use of information and knowledge.

### Measurement and reporting

Support decisions and improvement by reducing uncertainty.

### Organizational change management

Manage the human side of change so benefits last.

### Portfolio management

Secure the right mix of programmes, projects, products and services.

### Project management

Deliver all projects successfully.

### Relationship management

Establish and nurture links with stakeholders.

### Risk management

Understand and handle risk across the organisation.

### Service financial management

Use financial resources and investments effectively.

### Strategy management

Formulate goals and allocate resources to achieve them.

### Supplier management

Manage suppliers and their performance to support quality provision.

### Workforce and talent management

Right people, right skills, right roles.

### Learn the purposes

Foundation questions test purpose statements, not activity lists.

5.1.1

# Architecture management — purpose and problem

- The purpose is to provide an understanding of all the different elements that make up an organisation and how those elements interrelate, so the organisation can achieve its current and future objectives.
- It supplies the principles, standards and tools that let an organisation manage complex change in a structured and Agile way.
- As the modern environment and ecosystem have grown more complex, so have the challenges: efficiency and automation, but also managing complexity and achieving agility and resilience.
- Without the visibility and coordination architecture management provides, an organisation becomes a labyrinth of third-party contracts, variant processes across silos, needlessly customised products, and legacy infrastructure.
- The result is a landscape in which every change is harder to implement and carries far higher risk.

**SCOPE**

A complete practice addresses all architecture domains: business, service, information, technology and environment.

In a smaller, less complex organisation a single architect may develop one integrated architecture.

5.1.1

# The five architecture domains

1

## Business architecture

Views the organisation's capabilities against the detailed activities needed to create value, compares them with strategy, and performs a gap analysis of target state against current capability. Gaps are prioritised and closed incrementally; a roadmap describes the transformation.

2

## Service architecture

A view of all services provided, their interactions, and service models describing structure (how components fit together) and dynamics (activities, flow of resources, interactions). A service model can be a blueprint for several services.

3

## Information systems architecture

Describes the logical and physical data assets and the data management resources, and shows how information is managed and shared. Information has real, measurable value and underpins decisions, so it must be complete, accurate and accessible to those authorised.

4

## Technology architecture

Defines the software and hardware infrastructure needed to support the portfolio of products and services.

5

## Environmental architecture

Describes external factors affecting the organisation and the drivers for change, together with the types and levels of environmental control. It spans developmental, technological, business, operational, organisational, political, economic, legal, regulatory, ecological and social influences.

### Ghana lens

For a Ghanaian institution the environmental architecture is rarely theoretical: regulator directives, power supply, currency movement and connectivity all belong in it.

FIGURE 5.1

# Architecture management across the value chain

*Involved in all six activities, but most instrumental in plan, improve, and design and transition.*

### Plan • high

Develops and maintains the reference architecture describing current and target states for business, information, data, application, technology and environment. This is the basis for all planning.

### Improve • high

Many improvement opportunities are identified by reviewing the business, service, information, technical and environment architectures.

### Design and transition • high

Owns the service architecture used to evaluate whether an approved product or service still meets its investment objectives, and supplies the service model used as a blueprint.

### Engage

Supports understanding of the organisation's readiness to address new or under-served markets and a wider range of products, and to respond faster to changing circumstances.

### Obtain / build

The reference architectures make it clear which products, services or service components need to be obtained or built.

### Deliver and support

The reference architectures are used continually in the operation, restoration and maintenance of products and services.

5.1.2

# Continual improvement — purpose and scope

- The purpose is to align the organisation's practices and services with changing business needs through ongoing improvement of products, services and practices, or any element involved in managing them.
- Scope includes developing improvement-related methods and techniques, and propagating a continual improvement culture across the organisation in line with overall strategy.
- The commitment must be embedded in every fibre of the organisation. Where it is not, daily operational concerns and major project work will eclipse improvement work.
- Different types of improvement call for different methods: some are best run as a multi-phase project, others as a single quick effort.
- The continual improvement model from Chapter 4 applies to any type of improvement — from organisation-wide change down to a single configuration item.

**NOTE THE RELATIONSHIP**

Chapter 4 gave you the model and the improve value chain activity. This is the practice — the resources that do the work day to day.

5.1.2

# Key activities of the practice

1

## Encourage improvement

Promote continual improvement across the whole organisation.

2

## Secure time and budget

Improvement that is unfunded and unscheduled does not happen.

3

## Identify and log

Capture improvement opportunities wherever they arise.

4

## Assess and prioritise

Weigh opportunities against each other rather than acting first-come-first-served.

5

## Make the business case

Justify improvement action in terms decision-makers can act on.

6

## Plan and implement

Turn the approved case into executed change.

7

## Measure and evaluate

Establish whether the improvement actually delivered its intended result.

8

## Coordinate across the organisation

Prevent improvement efforts from colliding or duplicating each other.

5.1.2

# Methods, techniques and the culture that carries them

## Assessing the current state

Techniques include SWOT analysis, a balanced scorecard review, and internal or external assessments and audits — often several in combination. Organisations should build competence in the methods that suit their needs.

## Where approaches come from

Lean offers perspectives on eliminating waste; Agile focuses on incremental improvement at a cadence; DevOps works holistically so improvements are both well designed and effectively applied.

## Choose a few and cultivate them

Committing formally to too many approaches fragments the effort. A few well-understood methods give teams a shared way of working and therefore more change, faster.

## But stay open

Staff skilled in alternative methods should be encouraged to apply them where sensible; successful alternatives can join the repertoire, and older methods may be retired if newer ones perform better.

## Everyone's responsibility

A full-time team may lead the work, but participation is a core part of everyone's job — which is why contribution belongs in job descriptions, personal objectives and supplier contracts. Senior leadership must visibly own it or culture will not shift.

## Suppliers are inside the effort

Where third parties form part of the service landscape, contracts should state how they will measure, report on and improve their services — and what data they must supply for internal improvement.

5.1.2

# The continual improvement register

- A continual improvement register (CIR) is a database or structured document used to track and manage improvement ideas from identification through to final action.
- An organisation may hold more than one: registers can exist at individual, team, departmental, business unit and organisational levels, or one master register can be segmented by use and audience.
- Ideas often surface elsewhere first — during project execution or software development — and should then be documented for attention as part of ongoing improvement.
- As new ideas arrive, registers are used to reprioritise constantly, which makes work visible: what is in progress, what is complete, and what has been set aside for later.
- The structure and the name matter far less than the discipline: ideas must be captured, documented, assessed, prioritised and acted on.

**PRACTICES THAT FEED IT**

Problem management uncovers issues handled through improvement.

Organisational change management makes the resulting changes stick.

Project management organises and executes the larger initiatives.

FIGURE 5.2

# Continual improvement across the value chain

*The practice is integral to every other practice, to the whole service lifecycle, and to the SVS itself.*

### Improve • high

This is the practice's home activity. It structures the resources and activities that enable improvement at every level of the organisation and of the SVS itself.

### Plan

Applied to planning activities, methods and techniques so that they remain relevant to the organisation's current objectives and context.

### Engage, design and transition, obtain/build, deliver and support

Each of these activities is itself subject to continual improvement, and the practice is applied to all of them without exception.

# Information security management — purpose

- The purpose is to protect the information the organisation needs to conduct its business.
- That means understanding and managing risks to the confidentiality, integrity and availability of information.
- It also covers authentication — establishing that someone is who they claim to be — and non-repudiation, so that a person cannot deny having taken an action.
- The required security is established through policies, processes, behaviours, risk management and controls.
- It must be driven from the most senior level, based on clearly understood governance requirements and organisational policies; most organisations maintain a dedicated security team that performs risk assessments and defines policies, procedures and controls.

**THE THREE-WAY BALANCE**

Prevention — ensuring security incidents do not occur.

Detection — rapidly and reliably detecting what cannot be prevented.

Correction — recovering from incidents once detected.

5.1.3

# Security that helps rather than hinders

## Protection versus innovation

Controls that are too restrictive can do more harm than good, or simply be circumvented by people trying to work more easily. Controls should consider the whole organisation and align with its risk appetite.

## It touches every practice

Information security management creates controls that every other practice must consider when planning how work is done — and depends on those practices in turn to protect information.

## People decide the outcome

Well-trained staff who attend to policy help detect, prevent and correct incidents. Poorly trained or insufficiently motivated staff are themselves a major vulnerability.

## Supporting processes

- Information security incident management
- Risk management
- Control review and audit
- Identity and access management
- Event management

## Supporting procedures

- Penetration testing and vulnerability scanning
- Managing security-related changes, such as firewall configuration changes

## In high-velocity environments

Security is integrated as far as possible into the daily work of development and operations, shifting reliance from control of process towards verification of preconditions such as expertise and integrity.

FIGURE 5.3

# Information security management across the value chain

*Involved in every value chain activity — security is a property of the system, not a stage in it.*

## Plan

Security must be considered in all planning activity and built into every practice and every service.

## Improve

Considered in all improvement activity so that vulnerabilities are not introduced while making things better.

## Engage

Security requirements for new and changed services must be understood and captured. Every level of engagement, operational to strategic, must support security — and all stakeholders contribute, customers and suppliers included.

## Design and transition

Effective controls are designed and transitioned into operation. Security is considered alongside all other utility and warranty requirements.

## Obtain / build

Security is built into all components, based on the risk analysis, policies, procedures and controls defined by the practice — whether components are built internally or bought.

## Deliver and support

Detection and correction of security incidents is an integral part of this activity, not an adjunct to it.

5.1.4

# Knowledge management — purpose

- The purpose is to maintain and improve the effective, efficient and convenient use of information and knowledge across the organisation.
- Knowledge is one of an organisation's most valuable assets; the practice gives a structured approach to defining, building, re-using and sharing it in many forms — information, skills, practices, solutions and problems.
- As capture and sharing move towards digital solutions, the practice becomes more valuable rather than less.
- Knowledge is not simply information: it is the use of information in a particular context, understood with the user and the situation in mind.
- A 300-page manual is information; a short set of instructions that lets a service desk analyst find the answer quickly is knowledge that is fit for purpose.

**WHAT THE PRACTICE AIMS AT**

Stakeholders get the right information, in the proper format, at the right level and at the correct time — according to their access level and other relevant policies.

FIGURE 5.4

# Knowledge management across the value chain

*Acquisition matters too: developing, capturing and harvesting knowledge, whether formally documented or informal and tacit.*

### Deliver and support • high

Benefits most: re-use of solutions in standard situations, and better understanding of the context of non-standard situations that require analysis.

### Plan

Helps the organisation make sound portfolio decisions, define its strategy and other plans, and supports financial management.

### Improve

Improvement rests on understanding the current situation and trends, supported by historical information. Knowledge management supplies the context for assessing achievements and planning improvement.

### Engage

Relationships at every level depend on understanding their context and history; the practice helps the organisation understand its stakeholders.

### Design and transition

Knowledge of available solutions and technologies, and re-use of existing information, make this activity more effective.

### Obtain / build

Efficiency improves markedly with sufficient knowledge of available solutions and technologies and through re-use of information.

# Measurement and reporting — purpose

- The purpose is to support good decision-making and continual improvement by decreasing the level of uncertainty.
- This is achieved by collecting relevant data on managed objects and validly assessing that data in an appropriate context.
- Managed objects include products and services, practices and value chain activities, teams and individuals, suppliers and partners, and the organisation as a whole.
- Because those objects are connected, so are their metrics and indicators — which is why measurement starts from organisational goals rather than from available data.
- Goals may rest on profit, growth, competitive advantage, customer retention or public service; the relationship between high-level and subordinate goals must be made explicit.

**TWO DEFINITIONS**

Critical success factor (CSF): a necessary precondition for the achievement of intended results.

Key performance indicator (KPI): an important metric used to evaluate success in meeting an objective.

# KPIs, behaviour and reporting

## KPIs as motivator

Individual KPIs can work as a competitive motivator and drive positive results — provided they are set to meet clear business goals.

## The negative side

Target-setting for individuals can drive inappropriate behaviour when there is too much focus on individual KPIs. Service desk staff pushed to keep calls short may damage customer satisfaction and even resolution times.

## Set them for teams

Operational KPIs are better set at team level, allowing flexibility in targets and behaviour. Individuals still need guidance, but within team and organisational goals, and always in the context of value for the organisation.

## Reports and dashboards

Metrics are usually presented as reports or dashboards. Content must be relevant to the recipient and to the topic, and must make it easy to see what needs to be done and then act.

## The two questions a good report answers

- How far are we from our targets?
- What bottlenecks prevent us from achieving better results?

## Link back to Chapter 4

This is Goodhart's Law in practice: when a measure becomes a target, it stops being a good measure. The 'start where you are' principle warns against exactly this.

FIGURE 5.5

# Measurement and reporting across the value chain

### Plan

Enables strategy and service portfolio decisions by supplying detail on the current performance of products and services.

### Improve

Performance is constantly monitored and evaluated to support improvement, alignment and value creation.

### Engage

Engagement rests on correct, up-to-date and sufficient information, supplied as dashboards and reports.

### Design and transition

Provides information for management decisions at every stage before a service goes live.

### Obtain / build

Ensures transparency of development and procurement activity, enabling effective management and integration with the other activities.

### Deliver and support

Ongoing management of products and services depends on correct, current and sufficient performance information.

5.1.6

# Organisational change management — purpose

- The purpose is to ensure that changes in an organisation are implemented smoothly and successfully, and that lasting benefits are achieved by managing the human aspects of change.
- Improvements invariably require people to change how they work, how they behave, and sometimes their role entirely.
- Whether the change is to a practice, to the structure of the organisation, to technology, or is a new or changed service, people determine whether it succeeds.
- The practice aims to ensure that everyone affected accepts and supports the change — by removing or reducing resistance, addressing adverse impacts, and providing training, awareness and other support.
- It contributes to every part of the SVS wherever cooperation, participation and enthusiasm are required.

**CANNOT BE OUTSOURCED**

Accountability for organisational change management cannot be transferred to an external supplier. Execution may be delegated — even to suppliers — but leadership must come from inside the organisation.

# Four things that must be established and maintained

1

## Clear and relevant objectives

Objectives must make sense to stakeholders in the organisation's context, and the change must be seen to carry real value.

2

## Strong and committed leadership

The change needs active support from sponsors and day-to-day leaders. A sponsor is the manager or business leader who advocates and can authorise the change; leaders must visibly and consistently communicate their commitment.

3

## Willing and prepared participants

Change is made by willing participants. Willingness comes partly from conviction about its importance, and partly from feeling prepared through training, awareness and regular communication.

4

## Sustained improvement

Many changes fail because people revert to old ways after time passes. Value must be reinforced through regular communication, addressing impacts, and sponsor support — and the message is stronger when validated by metrics.

TABLE 5.2

# Activities of organisational change management

| Activity | Helps to deliver |
|---|---|
| **Creation of a sense of urgency** | Clear and relevant objectives; willing participants |
| **Stakeholder management** | Strong and committed participants |
| **Sponsor management** | Strong and committed leadership |
| **Communication** | Willing and prepared participants |
| **Empowerment** | Prepared participants |
| **Resistance management** | Willing participants |
| **Reinforcement** | Sustained improvement |

*Cultural difference matters: audiences respond to different messages and drivers, whether the difference arises from geography, nationality or corporate history.*

FIGURE 5.6

# Organisational change management across the value chain

*Its closest partners are continual improvement and project management; also measurement and reporting, workforce and talent, and relationship management.*

### Improve • high

Without proper organisational change management, improvement simply cannot be sustained.

### Plan

Decisions about change at portfolio level trigger the practice in support of an approved initiative.

### Engage

The practice actively engages stakeholders at every stage of a change, not only at announcement.

### Design and transition

Essential for deploying a new service or making a significant change to an existing one.

### Obtain / build

Ensures engagement and cooperation within projects and across them.

### Deliver and support

Continues during live operation to confirm that the change has actually been adopted and is being sustained.

5.1.7

# Portfolio management — purpose and activities

- The purpose is to ensure the organisation has the right mix of programmes, projects, products and services to execute its strategy within its funding and resource constraints.
- It is a coordinated collection of strategic decisions that together balance organisational change against business as usual.
- Activities include applying a systematic framework to define and deliver the portfolio; defining products and services and linking them to agreed outcomes; and evaluating and prioritising incoming proposals against resource constraints, existing commitments and strategy.
- It implements strategic investment appraisal based on value, cost, risk, resource constraints, inter-dependencies and impact on existing business activity.
- It then analyses and tracks investments, monitors overall portfolio performance, proposes adjustments as priorities shift, and reviews portfolios for progress, outcomes, cost, risk, benefit and strategic contribution.

**WHY IT MATTERS**

Portfolio management determines how resources are allocated, deployed and managed across the organisation — aligning resources and capabilities with customer outcomes as part of strategy execution.

5.1.7

# Three portfolios the practice keeps

1

## Product / service portfolio

The complete set of products and services managed by the organisation, representing its commitments and investments across all customers and market spaces — current contractual commitments, new development, and improvement plans arising from continual improvement. It may include third-party products that form part of the offering.

2

## Project portfolio

Manages and coordinates authorised projects so objectives are met within time and cost constraints and to specification. It prevents duplication, keeps projects within agreed scope, and confirms that resources are available — used for single projects and for large programmes alike.

3

## Customer portfolio

Maintained by the relationship management practice and fed into portfolio management. It records all the organisation's customers and represents the relationship manager's view of internal and external customers receiving products or services.

## How they connect

Portfolio management uses the customer portfolio to make sure the relationship between business outcomes, customers and services is well understood, documents those linkages, and validates them with customers through relationship management.

# Traditional and Agile portfolio management

## The traditional view

- Success has historically been judged by whether implementation finished on time, within budget, and delivered the required outputs, outcomes and benefits.
- Focused on top-down planning, with work laid out over long time periods.
- Organisations have often struggled to demonstrate a return on their investment in change.
- Growing recognition that true success also depends on whether it was the right initiative in the first place.

## The Agile view

- Increased focus on visualising strategic themes and reprioritising the portfolio swiftly.
- Increases workflow, reduces batch sizes of work, and controls the length of longer-term development queues.
- Takes the build–measure–learn cycle used by individual Agile teams and applies it organisation-wide.
- Teams work together, use modular design and share findings — shifting focus from executing an inflexible plan to delivering value.
- Communicates widely, shares knowledge, and breaks barriers between silos.

FIGURE 5.7

# Portfolio management across the value chain

### Plan • high

Supplies information on the status of projects, products and services in the pipeline or catalogue and the strategic objectives they were designed to meet — and reviews portfolios for progress, value creation, cost, risk, benefit and strategic contribution.

### Improve

Identifies opportunities to improve efficiency and increase collaboration, eliminates duplication between projects, and identifies and mitigates risk. Approved improvement initiatives may join the relevant portfolio.

### Engage

When opportunities or demand are identified, prioritisation decisions rest on the organisation's strategy plus risk assessment and resource availability.

### Design and transition, obtain/build, deliver and support

Portfolio management is responsible for ensuring products and services are clearly defined and linked to business outcomes, so that these activities stay aligned with value.

# Project management — purpose

- The purpose is to ensure that all projects in the organisation are successfully delivered — by planning, delegating, monitoring and maintaining control of every aspect, and by keeping the people involved motivated.
- Projects are one of the means by which significant change is introduced: temporary structures created to deliver one or more outputs according to an agreed business case.
- They may stand alone or form part of a larger programme of interrelated projects; even stand-alone projects should be considered within the organisation's project portfolio.
- Successful project management matters because the organisation must simultaneously maintain current operations, transform those operations to compete, and continually improve its products and services.
- That balance between projects and business as usual touches resources, service levels, customer relationships and productivity, so capacity and capability must be part of the approach.

**PEOPLE DECIDE IT**

The best project plan amounts to very little if the right people are not involved at the right time — and many team members are seconded from operations full- or part-time.

5.1.8

# Two delivery approaches

## Waterfall

- Works well where requirements are known up front and unlikely to change significantly.
- Suits environments where defining the work matters more than the speed of delivery.
- Gives predictability where predictability is genuinely available.
- Common where funding, procurement or regulation demands a fixed specification in advance.

## Agile

- Works best where requirements are uncertain and likely to evolve rapidly over time.
- Suits changing business needs and shifting priorities.
- Speed of delivery is often prioritised over precise definition of requirements.
- Requires funding and governance models that can tolerate an evolving scope.

FIGURE 5.8

# Project management across the value chain

### Obtain / build • high

Obtaining new resources, and development and integration work, is usually performed as a project, and the full range of project techniques applies here.

### Design and transition • high

The design of a practice or service can be run as a project or as an iteration within a larger project; the same applies to some transitions.

### Plan

Supports strategic and tactical planning with methods and tools.

### Improve

Many improvement initiatives are large and complex, which makes project management the relevant practice for managing them.

### Engage

Stakeholder engagement is central to delivering any project; the practice supplies stakeholder management tools and techniques.

### Deliver and support

Design, transition and handover to internal or external consumers must be planned and executed so that business as usual is not compromised.

5.1.9

# Relationship management — purpose and what it ensures

## Purpose

To establish and nurture the links between the organisation and its stakeholders at strategic and tactical levels — including the identification, analysis, monitoring and continual improvement of relationships with and between stakeholders.

### Needs understood

Stakeholders' needs and drivers are understood, and products and services are prioritised appropriately.

### Satisfaction maintained

Stakeholder satisfaction is high and a constructive relationship is established and maintained.

### Priorities articulated

Customers' priorities for new or changed products and services are established and articulated in line with desired business outcomes.

### Complaints handled

Complaints and escalations are handled well, through a sympathetic yet formal process.

### Value facilitated

Products and services facilitate value creation for consumers and for the organisation, and for all stakeholders in line with strategy and priorities.

### Conflicts mediated

Conflicting stakeholder requirements are mediated appropriately rather than left to escalate.

## Beyond the consumer

Providers naturally concentrate on sponsors, customers and users. The practice should apply to all relevant parties, internal and external — which is why it contributes to every value chain activity and to multiple value streams.

FIGURE 5.9

# Relationship management across the value chain

### Engage • high

This is the practice responsible for engaging with internal and external customers to understand their requirements and priorities.

### Plan

Supplies information on the requirements and expectations of internal and external customers, and assists strategic assessment and prioritisation across portfolios and market spaces.

### Improve

Seeks to harmonise different organisational relationships with internal and external customers so that targeted benefits are realised through improvement.

### Design and transition

Coordinates customer feedback as part of design, and ensures inconvenience and adverse impacts during transition are prevented or minimised.

### Obtain / build

Provides the customer requirements and priorities used to select what should be obtained or built.

### Deliver and support

Responsible for ensuring a high level of customer satisfaction and a constructive relationship are established and maintained.

5.1.10

# Risk management — purpose

- The purpose is to ensure that the organisation understands and effectively handles risk. Managing risk is essential to ongoing sustainability and to creating value for customers.
- Risk management is an integral part of all organisational activity and therefore central to the SVS.
- Risk is usually seen as something to avoid because of its association with threats — but risk is also associated with opportunity, and failing to take opportunities is itself a risk.
- The opportunity cost of under-served market spaces and unfulfilled demand is a risk to be avoided.
- Decisions must be balanced: the potential benefits should be worth more than the cost of addressing the risk. Innovation is inherently risky, yet can deliver major gains in products, competitive advantage, agility and resilience.

**PORTFOLIO AND RISK**

The organisation's portfolio maps to an underlying portfolio of risks. Where service management works, catalogue and pipeline represent opportunities; where it does not, the same items represent threats through failure, demand patterns, commitments and cost.

# For risk management to be effective, risks must be...

1

## Identified

Uncertainties that would affect the achievement of objectives within a particular organisational activity must be considered and then described, so that there is common understanding of what is actually at stake.

2

## Assessed

The probability, impact and proximity of individual risks must be estimated so they can be prioritised, and so that the overall level of risk — the risk exposure — associated with the activity is understood.

3

## Treated

Appropriate responses must be planned, with owners and actionees assigned, and then implemented, monitored and controlled. A response that is planned but never monitored is not a treatment.

# Three principles, and the behaviours that carry them

1

## Risk is part of business

Not all risks are to be avoided — risk-taking is required for long-term sustainability. Risks must be identified, understood and assessed against the risk appetite, then managed and monitored.

2

## Consistency across the organisation

The practice must be managed holistically for consistency, with ongoing stakeholder consultation and enough flexibility for tailored procedures where organisational units or customers require them.

3

## Culture and behaviour matter

Appropriate culture and behaviour at all levels must be embedded as part of 'the way we do things' — proactive behaviour, transparency of procedures and accountabilities, and active encouragement of reporting.

### Behaviours to look for

- Understanding that risk management underpins sustainability and business goals
- Transparency and clarity of roles, responsibilities and accountabilities
- Following up reports of risks, incidents and opportunities
- Remuneration that neither discourages incident reporting nor rewards over-reporting
- Learning and growth in maturity from the organisation's and others' experience

### ISO 31000:2018

Provides a general perspective on the purpose and principles of risk management, applicable at all levels in any organisation. It frames the purpose of risk management as the creation and protection of value, and states that it improves performance, encourages innovation and supports the achievement of objectives.

FIGURE 5.10

# Risk management across the value chain

*Plan is where risk management concentrates: it supplies essential input on the risks that drive variability of outcomes.*

## Plan • high

- Shifts in customer demand and priorities
- Legal and regulatory changes
- Competitors
- Dependencies on suppliers and partners
- Technological changes
- Conflicting stakeholder requirements

## Improve

All improvement initiatives should be assessed and continually controlled by risk management, which supplies an important perspective for prioritisation, planning and review.

## Engage

Helps identify key stakeholders and optimise engagement using information such as risk appetite and risk profiles.

## Design and transition

Products and services should be designed to address prioritised risks and be scalable as demand changes. New or changed services carry varying risk that must be identified and assessed before approval, then managed as part of the change.

## Obtain / build

Should inform decisions about obtaining or building products, services or service components.

## Deliver and support

Helps ensure ongoing delivery is maintained at the agreed level and that events are managed according to the risks they introduce.

# Service financial management — purpose and activities

*To work inside the SVS, the practice must align with portfolio management, project management and relationship management.*

## Purpose

To support the organisation's strategies and plans for service management by ensuring that its financial resources and investments are used effectively. It supports decisions by the governing body and management about where best to allocate financial resources, and gives visibility into budgeting, costing and accounting for products and services.

## Budgeting and costing

Predicting and controlling income and expenditure. Budgeting is a periodic negotiation cycle plus ongoing monitoring, capturing forecast and actual demand and translating it into operating and project costs, budgets and rates. Service-based budgeting builds funding models on the full cost of providing or consuming a service.

## Accounting

Enables the organisation to account fully for how money is spent, comparing forecast against actual cost and expenditure — particularly identifying usage and cost by customer, service and activity or cost centre. It involves ledgers, charts of accounts and journals.

## Charging

Formally invoicing service consumers, usually external ones, for services provided. Charging is optional — but every service requires a funding model, because all costs must be adequately funded by an agreed method.

# How technology is reshaping financial management

## Digital technologies

Financial institutions use cloud, big data, analytics and AI to gain or maintain competitive advantage. New entrants operate without legacy IT, technical debt or bureaucratic process, and therefore tend to be more Agile. Deeper customer insight from analytics is itself generating new products and services.

## Blockchain

Enabled through cloud-based services and originally developed for decentralised management of crypto-currencies, allowing transactions to be audited and verified automatically and cheaply. Distributed ledgers record transactions across many computers, so a record cannot change without altering all subsequent blocks and without network consensus.

## From CAPEX to OPEX

IT resources were traditionally obtained through upfront capital expenditure. Under the cloud model, infrastructure, platforms and software arrive 'as a service' on subscription or pay-as-you-use terms, funded from operational expenditure instead.

## Flexible budgets

Scaling cloud services in an Agile, on-demand way requires flexible IT budgets. Fixed budgets forecast months in advance struggle to account for that kind of scaling.

## Procurement rules must change

Cloud services are generally sold on a variable price: the more you consume, the more you pay. Organisations that cannot buy variable-priced IT erect a self-made barrier to using cloud services at all.

## Ghana lens

Where procurement thresholds and approval cycles are set by statute, this is not a policy preference but a structural constraint — and one worth naming explicitly in a business case.

FIGURE 5.11

# Service financial management across the value chain

### Plan • high

Plans at every level need funding based on information, including financial information. The practice supports planning with budgets, reports and forecasts.

### Improve

All improvements should be prioritised with return on investment in mind; the practice supplies the tools and information for evaluating and prioritising them.

### Engage

Financial considerations shape service relationships with consumers, suppliers and partners — and for investors and sponsors the financial aspect may be the most important one.

### Design and transition

Keeps the activity cost-effective through financial planning and control, and ensures transparency of product and service costs by accounting for design and transition spend.

### Obtain / build

Obtaining resources of all types is supported by budgeting, to ensure sufficient funding, and accounting, to ensure transparency and evaluation.

### Deliver and support

Ongoing operational costs are a major expenditure and, in commercial organisations, ongoing delivery is also the source of income. Charging, where applicable, supports billing and reporting.

# Strategy management — purpose and objectives

- The purpose is to formulate the goals of the organisation and adopt the courses of action and allocation of resources needed to achieve them.
- Strategy management establishes direction, focuses effort, defines or clarifies priorities, and provides consistency or guidance in response to the environment.
- The starting point is understanding the context of the organisation and defining the desired outcomes; strategy then supplies the criteria and mechanisms for prioritising resources, capabilities and investment.
- Objectives include analysing the environment for opportunities; identifying constraints and how to remove or reduce them; and agreeing perspective and direction with stakeholders, including vision, mission and principles.
- They also include establishing position relative to customers and competitors, translating strategy into tactical and operational plans for each unit, coordinating execution across all three levels, and managing changes to the strategy itself.

**WHOSE JOB IS IT?**

Traditionally the responsibility of senior management and the governing body — setting objectives, specifying how they will be met, and prioritising the investment required.

# Strategy in a fast-changing environment

*Contribution across the value chain: heaviest in plan, where strategy becomes tactical and operational plans for each unit.*

## Strategy is becoming more fluid

Traditional practice rested on careful deliberation, extensive research and scenario planning. The emphasis is shifting towards establishing the essential purpose and principles of the organisation, which can guide action even as circumstances change.

## A Lean strategy process

Can be used to balance the extremes of rigid planning on one side and uncontrolled experimentation on the other.

## Strategy as screen and yardstick

It provides overall direction and alignment, acting both as the screen that innovative ideas must pass and as the basis for evaluating the success of the SVS — encouraging creativity while keeping it in harmony with the organisation.

## Strategy must enable value creation

A good business model describes how objectives will be fulfilled. Strategy should include a way to make services and products uniquely valuable to customers, and must define the approach for delivering better value.

## Not only for large organisations

Smaller organisations need a clear perspective, positioning and plans just as much, so that they remain relevant to their customers.

## A high-performance strategy

Enables an organisation to outperform competing alternatives consistently — over time, across business cycles, through industry disruption, and when leadership changes.

# Supplier management — purpose and central activities

- The purpose is to ensure that the organisation's suppliers and their performance are managed appropriately, to support the seamless provision of quality products and services.
- It includes building closer, more collaborative relationships with key suppliers, to uncover and realise new value and to reduce the risk of failure.
- Creating a single point of visibility and control ensures consistency across all products, services, components and procedures provided by internal and external suppliers — including customers acting as suppliers.
- The practice maintains a supplier strategy, policy and contract management information, and negotiates and agrees contracts aligned with business needs and service targets, often through legal, procurement or commercial functions.
- It manages relationships and contracts through planning, design, build, orchestration, transition and operation, and monitors supplier performance against contractual terms while seeking better value for money.

**INTERNAL SUPPLIERS COUNT**

Where the supplier is internal, there still needs to be an internal agreement. The absence of a contract is not the absence of an obligation.

# Four sourcing relationships

1

## Insourcing

Products or services are developed and delivered internally by the organisation itself.

2

## Outsourcing

External suppliers provide products and services previously provided internally. It involves substitution — the replacement of internal capability by the supplier's.

3

## Single source or partnership

Procurement from one supplier: either a single supplier delivering everything directly, or an external service integrator managing all supplier relationships on the organisation's behalf. The mutual interdependence fosters quality, reliability, short lead times and cooperation.

4

## Multi-sourcing

Procurement from more than one independent supplier, whose products and services can be combined into new services. As organisations pursue specialisation to increase agility, this is increasingly preferred — with a move towards internal service integration capability or an external integrator.

## The sourcing strategy

Sometimes called the supplier strategy, it defines how the organisation will leverage suppliers in achieving its overall service management strategy. A successful one requires thorough understanding of objectives, the resources needed, market factors, and the risks of each approach.

## Small suppliers, large effect

Individual suppliers may play a minor and indirect role in value generation, yet collectively make a direct and important contribution to value and to strategy execution.

# Selecting suppliers, and managing them afterwards

## Select on three criteria

- Importance and impact — the value of the supplier's service to the business
- Risk — the risks associated with using the service
- Cost — the cost of the service and of its provision

## Other factors that decide it

The supplier's willingness or ability to customise its offering or work cooperatively in a multi-supplier environment; how much influence the organisation or integrator has over the supplier's performance; and how dependent one supplier is on others.

## Supplier planning

Understand new or changed service requirements and review enterprise documentation to develop the sourcing strategy and supplier management plan — with business analysis, portfolio management, service design and service level management.

## Evaluation and negotiation

Identify, evaluate and select suppliers for new or changed business services; then develop, negotiate, review, update, finalise and award contracts. Failed negotiation triggers a new contract, an update, or termination.

## Categorisation and management

Suppliers are categorised periodically and after contract awards — commonly as strategic, tactical or commodity. Contract management then ensures value for money and delivery of agreed performance against targets.

## Warranty, performance, renewal

Warranty clauses are managed and claims made when issues arise. Operational measures agreed with suppliers are tracked continuously and consolidated on a scorecard, revealing systemic problems. Renewals and terminations follow specific or periodic performance reviews.

5.1.13.4

# Service integration

- Service integration coordinates or orchestrates all the suppliers involved in developing and delivering products and services.
- It focuses on end-to-end provision: controlling all interfaces and outcomes from suppliers, and facilitating collaboration between them.
- An organisation may act as its own service integrator, use a third-party integrator, or build a hybrid in which it keeps part of the function and augments the rest externally. The function can also be operated by a lead supplier.
- The integrator is also responsible for assurance — performance management and reporting, defining roles and responsibilities, and maintaining relationships across all parties.
- It chairs the regular forums and steering committees where issues are addressed, priorities agreed and decisions taken.

**VALUE CHAIN CONTRIBUTION**

Heaviest in engage — the practice responsible for engaging with all suppliers, evaluating and selecting them, negotiating agreements and managing the relationships.

Also: sourcing strategy for plan; contract requirements for design and transition; procurement support for obtain/build; live performance for deliver and support.

# Workforce and talent management — purpose

- The purpose is to ensure the organisation has the right people, with the appropriate skills and knowledge, in the correct roles to support its business objectives.
- The practice covers planning, recruitment, onboarding, learning and development, performance measurement and succession planning.
- It plays a critical role in establishing organisational velocity, by helping the organisation understand and forecast future demand for services and ensuring the right competencies are available at the right time.
- Achieving this reduces backlogs, improves quality, avoids rework caused by defects, reduces wait time, and closes knowledge and skills gaps.
- As third-party use and automation of repeatable work increase, traditional roles are changing dramatically — which is why this practice belongs to leaders and managers at every level, not to one department.

**WHAT A GOOD PEOPLE STRATEGY DOES**

Identifies roles and the associated knowledge, skills and attitudes needed to run the organisation day to day — and addresses the emerging technologies and leadership and change capabilities needed for future growth.

# Definitions to know

## Organizational velocity

The speed, effectiveness and efficiency with which an organisation operates. It influences time to market, quality, safety, cost and risk.

## Competencies

The combination of observable and measurable knowledge, skills, abilities and attitudes that contribute to enhanced employee performance and, ultimately, organisational success.

## Skills

A developed proficiency or dexterity in thought, verbal communication or physical action.

## Ability

The power or aptitude to perform physical or mental activities related to a profession or trade.

## Knowledge

The understanding of facts or information acquired through experience or education; the theoretical or practical understanding of a subject.

## Attitude

A set of emotions, beliefs and behaviours towards a particular object, person, thing or event.

FIGURE 5.14

# Workforce and talent management activities




### Workforce planning

Translating strategy and objectives into desired organisational capabilities, then into competencies and roles.

### Recruitment

Acquiring new employees and contractors to fill identified gaps in desired capabilities.

### Performance measurement

Regular measurement and assessment against established job roles, based on pre-defined competencies.

### Personal development

The employee's own use of published job roles and competency frameworks to plan growth and advancement.

### Learning and development

Targeted education and experiential learning using formal and non-formal methods.

### Mentoring and succession planning

Formal mentoring, engagement and succession planning provided by leadership.

FIGURE 5.15

# Workforce and talent management across the value chain

### Plan • high

Workforce planning is a specific output of this activity, as leadership evaluates current organisational capability against future requirements for resources and for the products and services in the portfolio.

### Improve • high

All improvements require sufficiently skilled and motivated people; the practice ensures those requirements are understood and met.

### Engage

Works with relationship management, service request management and the service desk to understand and forecast changing demand, and how it will direct workforce planning and talent activities.

### Design and transition

Specific focus on the knowledge, skills and abilities related to systems thinking and design thinking.

### Obtain / build

Focus on knowledge, skills and abilities related to collaboration, customer focus, quality, speed and cost management.

### Deliver and support

Focus on knowledge, skills and abilities related to customer service, performance management, and customer interactions and relationships.

CONSOLIDATION

# Where each general practice does most of its work

*A revision aid, not an examinable grid: ● marks the value chain activities where each practice concentrates.*

| General management practice | Plan | Improve | Engage | Design & transition | Obtain / build | Deliver & support |
|---|---|---|---|---|---|---|
| Architecture management | ● | ● | · | ● | · | · |
| Continual improvement | · | ● | · | · | · | · |
| Information security management | ● | · | · | ● | ● | ● |
| Knowledge management | · | · | · | · | · | ● |
| Measurement and reporting | ● | ● | · | · | · | ● |
| Organizational change management | · | ● | ● | ● | · | · |
| Portfolio management | ● | · | · | · | · | · |
| Project management | · | · | · | ● | ● | · |
| Relationship management | · | · | ● | · | · | ● |
| Risk management | ● | ● | · | ● | · | · |
| Service financial management | ● | · | · | · | · | ● |
| Strategy management | ● | · | · | · | · | · |
| Supplier management | ● | · | ● | · | ● | · |
| Workforce and talent management | ● | ● | · | · | · | · |

B

CONTEMPORARY ISSUES

# The general practices under current conditions

What has changed since ITIL 4 was published — and what it means for organisations in Ghana.

CONTEMPORARY ISSUE 1

# Artificial intelligence in the general practices

## Knowledge management first

Retrieval-augmented assistants change the economics of the knowledge base: the constraint moves from finding content to trusting it. Retiring stale knowledge becomes more urgent, not less, because a model will confidently repeat it.

## Measurement and reporting

Natural-language querying of operational data lowers the barrier to analysis — and raises the risk that a plausible number goes unchallenged. Provenance becomes a reporting requirement.

## Workforce and talent

Automation of repeatable work shifts demand towards supplier management, data literacy and judgement. Workforce planning has to forecast roles that did not exist in the last plan.

## Risk and information security

Model behaviour, data provenance, prompt injection and third-party model suppliers all become entries in the risk register and in security policy.

## Architecture management

Where models are hosted, what data crosses which boundary, and which systems may call them are architecture decisions, not procurement details.

## For discussion

If an AI assistant drafts a service report that a manager acts on, which practice owns the accuracy of that report?

CONTEMPORARY ISSUE 2

# FinOps and the economics of consumption

## Cost has become an operational signal

Under consumption pricing, cost is generated daily by engineering decisions rather than annually by a procurement event. Service financial management therefore needs near-real-time visibility.

## Showback and chargeback

Attributing cloud spend to a service, team or customer makes the accounting activity concrete — and often reveals which services are genuinely funded and which are quietly subsidised.

## Budgeting under exchange-rate exposure

For Ghanaian organisations, cloud services priced in foreign currency turn a technical decision into a treasury one. Budget assumptions should state the rate they rest on.

## Procurement rules as the binding constraint

Where statute requires a fixed price and a fixed quantity, variable-priced consumption services are difficult to buy at all. This is a policy problem before it is a technical one.

## Portfolio consequences

When the true running cost of each service becomes visible, portfolio decisions change: some services turn out not to justify their consumption.

## For discussion

Which activity of service financial management would most improve decisions in your organisation if it were done properly?

CONTEMPORARY ISSUE 3

# Cyber resilience, privacy and regulation

## From security to resilience

Ransomware and third-party compromise have shifted the emphasis from preventing incidents to continuing to operate through them — prevention, detection and correction weighted differently than a decade ago.

## Data protection in Ghana

The Data Protection Act, 2012 (Act 843) and the Data Protection Commission impose obligations on registration, lawful processing and data subject rights that shape what information security management must control.

## Cybersecurity oversight

The Cybersecurity Act, 2020 (Act 1038) established the Cyber Security Authority, with licensing and incident reporting expectations for providers of relevant services.

## Supplier management carries it

Obligations follow the data into the supply chain. Contract requirements — an output of several value chain activities — are where security expectations are actually enforced.

## Security that people can follow

The warning in the practice holds: controls that are too restrictive get circumvented. Usable controls beat strict controls that staff route around.

## For discussion

Which supporting process of information security management is weakest in the organisations you know — and what does that cost?

CONTEMPORARY ISSUE 4

# Sourcing, integration and the ecosystem

## Multi-sourcing is now the default

Specialisation has made single-supplier delivery rare. The practical question is no longer whether to multi-source but who integrates the result.

## Service integration as a capability

Whether the integrator is internal, external, hybrid or a lead supplier, someone must own end-to-end assurance, interfaces and the forums where decisions get made.

## The Ghanaian ecosystem

Telcos, payment switches, systems integrators and cloud resellers appear in most service value streams. One organisation's supplier management is another's relationship management.

## Concentration risk

When many organisations depend on the same few providers of connectivity or payment rails, an individually rational sourcing choice becomes a shared systemic exposure.

## Contracts as improvement instruments

Continual improvement asks that supplier contracts specify how services will be measured, reported on and improved — a clause that is easy to write and rarely written.

## For discussion

Name a service you rely on where the integrator role is genuinely unowned. What happens when it fails?

CONTEMPORARY ISSUE 5

# People, change and the Ghanaian workplace

### The skills market is competitive

Cloud, security and data skills are internationally tradeable, and remote work has widened the market for Ghanaian talent. Retention has become a workforce planning variable, not an HR afterthought.

### Succession planning is thin

Where critical knowledge sits with one or two individuals, the organisation carries an unrecorded risk. Mentoring and succession planning are the treatment.

### Change fatigue is real

Successive system rollouts without reinforcement teach staff that new systems can be outlasted. That is a failure of the reinforcement activity, not of the technology.

### Culture is not a footnote

The practice explicitly requires cultural difference to be considered — geography, nationality, corporate history. In a multi-site Ghanaian organisation this is a design input for communication.

### Hybrid work and visibility

Distributed teams make work in progress harder to see, which raises the value of measurement, reporting and deliberate communication.

### For discussion

Which of the seven organisational change management activities would most improve the last major change you witnessed?

EXAMINATION FOCUS

# Six misconceptions to avoid

1

## A practice is not a process

It is a set of organisational resources — people, tools, information and processes together — shaped by the four dimensions.

2

## 'General' does not mean 'less important'

The label describes origin — adopted from general business management — not priority or difficulty.

3

## Organisational change management is not change control

One manages the human side of change; the other, a service management practice, authorises changes to services.

4

## Charging is optional; funding is not

Every service needs a funding model even where no invoice is ever raised.

5

## Risk management is not risk avoidance

Risk-taking is required for long-term sustainability; the practice manages exposure against appetite.

6

## Continual improvement is not a department

A small team may lead it, but participation belongs in every job description and every supplier contract.

SUMMARY

# What to take away from this session

- A management practice is a set of organisational resources designed for performing work or accomplishing an objective; the SVS holds 34 of them across three families.
- General practices came from general business management, service practices from the ITSM industry, and technical practices from technology management domains.
- High-velocity service delivery is reshaping all three families — and demands organisational change of significant scale where a legacy of low-velocity habits exists.
- The fourteen general practices set direction (strategy, portfolio, architecture, project), protect (risk, information security), inform (measurement, knowledge, continual improvement), and sustain (finance, suppliers, relationships, change, people).
- Every practice contributes to every value chain activity, but each concentrates somewhere — and knowing where is how you reason about a scenario question.

NEXT SESSION

Chapter 5 continues with the 17 service management practices and the 3 technical management practices.

ASSESSMENT PRACTICE

# Review questions

1

## Recall

How many general, service and technical management practices does the ITIL SVS contain, and where did each family originate?

2

## Define

State the purpose of the architecture management practice and name the five architecture domains.

3

## Explain

What is a continual improvement register, and why may an organisation hold more than one?

4

## Distinguish

Differentiate prevention, detection and correction in information security management.

5

## Analyse

Why should operational KPIs usually be set for teams rather than individuals? Give a service desk example.

6

## Apply

A ministry wants to move a system to the cloud but can only issue fixed-price contracts. Which practices are implicated, and what would you advise?

7

## Evaluate

Why can accountability for organisational change management not be outsourced, even when execution is delegated?

8

## Compare

Contrast single sourcing and multi-sourcing, and explain what service integration adds in each case.

TUTORIAL

# Group activity: a practice health check

## The brief

In groups of four, choose one organisation you know well — an employer, a bank, a ministry, a university department. You will assess three of its general management practices and report back.

## Step 1 — Choose and justify (10 min)

- Pick three of the fourteen practices
- Say why these three matter most for that organisation
- State the purpose of each in your own words

## Step 2 — Assess (20 min)

- For each: what exists today?
- Which value chain activity does it actually serve?
- Where is the largest gap between purpose and practice?

## Step 3 — Recommend (20 min)

- One improvement per practice
- Name the practice that must support it
- Say how you would measure success

## Step 4 — Present (10 min)

- Five minutes per group
- One slide or flip chart
- Expect to be challenged on your evidence

## Assessment criteria

- Correct use of practice terminology
- Realistic, specific recommendations
- Explicit link to a value chain activity
- Quality of the proposed measure

REFERENCE

# Key terms from this session

### Management practice

A set of organisational resources designed for performing work or accomplishing an objective.

### Organizational velocity

The speed, effectiveness and efficiency with which an organisation operates.

### Continual improvement register

A database or structured document tracking improvement ideas from identification to final action.

### Critical success factor

A necessary precondition for the achievement of intended results.

### Key performance indicator

An important metric used to evaluate success in meeting an objective.

### Sponsor

A manager or business leader who advocates a change and can authorise it.

### Risk appetite

The level of risk an organisation is willing to take.

### Service integration

Coordinating all suppliers involved in developing and delivering products and services, end to end.

### Multi-sourcing

Procurement of a product or service from more than one independent supplier.

### Competencies

Observable, measurable knowledge, skills, abilities and attitudes that improve performance.

### Service-based budgeting

Funding models built on the full cost of providing or consuming a service.

### Bi-modal IT

High-velocity service management operating alongside traditional practices.

SOURCES

# References and further reading

*Verify the currency of all legal and regulatory references before citing them in assessed work.*

## Primary text

- AXELOS (2019). ITIL® Foundation: ITIL 4 Edition. London: TSO — Chapter 5 introduction and sections 5.1 to 5.1.14.1.
- These slides paraphrase and summarise that material for classroom use; read the original text alongside them.
- ITIL® is a registered trade mark of AXELOS Limited.

## Supporting reading

- ISO 31000:2018 — Risk management guidelines.
- ISO/IEC 27001 — Information security management systems.
- AXELOS. ITIL 4: Direct, Plan and Improve.
- Kotter, J. P. Leading Change — the sense of urgency and reinforcement.
- FinOps Foundation. FinOps Framework — cloud cost management.

## Ghanaian context

- Data Protection Act, 2012 (Act 843).
- Cybersecurity Act, 2020 (Act 1038).
- Public Procurement Act, 2003 (Act 663) as amended.
- Bank of Ghana directives on outsourcing and operational resilience.

?

END OF CHAPTER 5, PART A

# Questions and discussion

Next: the 17 service management practices and the 3 technical management practices.

University of Professional Studies, Accra

UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA

ITIL 4 Foundation

# ITIL Management Practices

Chapter 5, Sections 5.2 – 5.2.17.2 | The seventeen service management practices

**Lecture deck** | 66 slides | Learning objectives, practice-by-practice breakdown, contemporary issues and Ghanaian public-sector applications

Faculty of Information Technology and Communication Studies • Department of Information Technology Studies

SESSION ROADMAP

# How this session is organised

1 **Foundations**

What a practice is, how the 34 practices are grouped, and how to read a value-chain heat map.

2 **Practices 5.2.1 – 5.2.4**

Availability, business analysis, capacity and performance, change control.

3 **Practices 5.2.5 – 5.2.9**

Incident, IT asset, monitoring and event, problem, release management.

4 **Practices 5.2.10 – 5.2.13**

Service catalogue, configuration, continuity and service design.

5 **Practices 5.2.14 – 5.2.17**

Service desk, service level, service request, validation and testing.

6 **Beyond the syllabus**

AIOps, FinOps, SRE, platform engineering, and Ghanaian public-sector practice.

BY THE END OF THIS SESSION YOU SHOULD BE ABLE TO

# Learning objectives (1 of 2)

1 **Explain the purpose of each practice**

State, in your own words, the purpose of all seventeen service management practices covered in section 5.2.

2 **Recall the examinable definitions**

Define availability, performance, change, incident, problem, known error, workaround, event, IT asset, configuration item, release, service request, service level and service level agreement.

3 **Distinguish closely related concepts**

Separate incident from problem, change control from organisational change management, and service continuity from major incident management.

4 **Map practices to the service value chain**

Identify which value chain activities each practice contributes to most heavily, and justify why.

BY THE END OF THIS SESSION YOU SHOULD BE ABLE TO

# Learning objectives (2 of 2)

### 5 Apply the practices to a real organisation

Given a scenario, recommend which practice or combination of practices addresses the problem described.

### 6 Evaluate measurement choices

Critique availability percentages, single-system SLA targets and other metrics that look healthy but hide poor customer experience.

### 7 Relate the practices to Ghanaian institutions

Illustrate each practice using public-sector and financial-sector systems familiar in Ghana, such as GIFMIS, GhIPSS and mobile money platforms.

### 8 Discuss contemporary developments

Explain how AIOps, cloud FinOps, site reliability engineering and platform engineering are reshaping these practices.

ORIENTATION

# Where section 5.2 sits in the ITIL 4 architecture

**RECALL FROM CHAPTER 4**

The service value system converts demand into value. Inside it, the service value chain provides six activities — plan, improve, engage, design and transition, obtain/build, deliver and support. Practices are the organisational resources that the value chain draws on to actually do the work.

1

### Guiding principles

Seven recommendations that shape every decision

2

### Governance

Direction, evaluation and oversight by the governing body

3

### Service value chain

Six interconnected activities that create the service

4

### Practices

34 sets of organisational resources — our focus today

5

### Continual improvement

Ongoing refinement across the whole system

GH

A practice is not a process. It bundles people, workflows, tools, information and supplier arrangements — the same four dimensions you met in chapter 3.

CATEGORIES

# The thirty-four ITIL practices at a glance

## 14

### General management

Adapted from general business management. Includes strategy management, portfolio management, architecture management, risk management, information security management, knowledge management, measurement and reporting, project management, relationship management, supplier management, workforce and talent management, continual improvement, service financial management and organisational change management.

## 17

### Service management

Developed within service management and IT service management. These are the seventeen practices covered in section 5.2 and examined in detail in this session.

## 3

### Technical management

Adapted from technology management for service management purposes: deployment management, infrastructure and platform management, and software development and management.

GH

Examination note: you are expected to recall the purpose of all 34, but only a subset in detail. Section 5.2 gives you the detailed seventeen.

OUR MAP FOR THE SESSION

# The seventeen service management practices

- 5.1 Availability management
- 5.2 Business analysis
- 5.3 Capacity and performance management
- 5.4 Change control
- 5.5 Incident management
- 5.6 IT asset management
- 5.7 Monitoring and event management
- 5.8 Problem management
- 5.9 Release management
- 5.10 Service catalogue management
- 5.11 Service configuration management
- 5.12 Service continuity management
- 5.13 Service design
- 5.14 Service desk
- 5.15 Service level management
- 5.16 Service request management
- 5.17 Service validation and testing

*Numbering shown is 5.2.1 to 5.2.17 in the manual, abbreviated here for space.*

METHOD

# How to read a value-chain heat map

Each practice in the manual is followed by a heat map showing how strongly it contributes to each of the six value chain activities. The shading runs on a four-point scale.

| 0 | 1 | 2 | 3 |
|---|---|---|---|
| No contribution | Low | Medium | High |

### What the shading is telling you

- Where the practice does most of its work day to day
- Which teams therefore need to own or resource it
- Which value chain activity suffers first when the practice is weak

### What it is not telling you

- It is not a maturity score for your organisation
- It is not a priority ranking between practices
- It is illustrative — your own context may differ legitimately

PART ONE

# Availability, analysis, capacity and change

- 5.2.1 Availability management
- 5.2.2 Business analysis
- 5.2.3 Capacity and performance management
- 5.2.4 Change control

5.2.1

# Availability management

KEY MESSAGE

The purpose of the practice is to make sure services deliver the levels of availability that have been agreed, so that the needs of customers and users are actually met.

DEFINITION — AVAILABILITY

The capability of an IT service, or any other configuration item, to carry out its agreed function at the moment it is needed.

## Note the wording

Agreed function, when required. Availability is judged against an agreement and against a moment of need, not against a raw uptime figure.

## Core activities of the practice

- Negotiating availability targets that the organisation can realistically achieve
- Designing infrastructure and applications capable of delivering those levels
- Ensuring services and components can generate the data needed to measure availability
- Monitoring, analysing and reporting on availability, and planning improvements to it

5.2.1

# Two ways to think about failure: MTBF and MTRS

Availability depends on two things — how often a service fails, and how quickly it comes back.

## Mean time between failures (MTBF)

- Measures how frequently the service fails
- A service with an MTBF of four weeks fails roughly thirteen times a year
- Raising MTBF means building components that break less often
- Still the priority where even a brief outage is catastrophic

## Mean time to restore service (MTRS)

- Measures how quickly service is restored after a failure
- An MTRS of four hours is an average across many incidents, not a guarantee
- Lowering MTRS means designing for fast, often automatic, recovery
- The dominant modern emphasis in service design

### The design shift

Older services were engineered for very high MTBF. The modern emphasis is on minimising MTRS through anti-fragile design — solutions that recover automatically and almost instantly, with negligible business impact.

5.2.1

# Measuring availability in ways customers recognise

Percentage availability calculated from MTBF and MTRS rarely matches what customers actually experienced. Four alternatives work better for particular service types.

### 1 User outage minutes

Incident duration multiplied by users affected, or the sum of minutes lost per user. Suits services that support productivity directly, such as email.

### 2 Number of lost transactions

Expected transactions in the period minus those that actually completed. Suits transaction-driven processes such as manufacturing or payments.

### 3 Lost business value

The productivity cost of the failures. Easy for customers to grasp and useful for investment cases, but hard to attribute cleanly to IT causes.

### 4 User satisfaction

Availability is among the most visible service characteristics, so satisfaction must be checked alongside formally agreed targets.

**GH** Applied locally: for a mobile money platform, lost transactions during peak market hours communicates far more to a business sponsor than a 99.6 per cent uptime figure.

5.2.1

# Who does availability management, and where it lands

## Organisational reality

- Most organisations have no dedicated availability management staff
- Activities are distributed — sometimes folded into risk management
- Elsewhere merged with continuity, or with capacity and performance
- Some organisations appoint site reliability engineers for specific products
- Regular testing of failover and recovery mechanisms is essential

## Culture beats procedure

Many organisations have a process for calculating and reporting availability metrics. But the manual is blunt about this: availability management is driven as much by culture, experience and knowledge as by following documented procedures. A perfectly written procedure will not save a team that has never rehearsed a failover.

## Contribution to the service value chain — involved in all six activities

- Plan: shapes service portfolio decisions and target setting • Improve: ensures improvements do not degrade what already works
- Engage: captures availability requirements • Design and transition: designs to target and tests availability controls
- Obtain/build: a criterion when building or sourcing components • Deliver and support: measurement and event response

5.2.2

# Business analysis

KEY MESSAGE

The purpose is to analyse a business, or an element of it, define the needs that emerge, and recommend solutions that address those needs or solve a business problem — in a way that facilitates value creation for stakeholders. It lets an organisation express what it needs, articulate why change is justified, and describe solutions aligned to its objectives.

## Key activities

- Analysing systems, processes, services and architectures in a shifting context
- Identifying and prioritising parts of the SVS needing improvement or innovation
- Proposing actions — not only system changes, but process, structural and staffing changes
- Documenting business requirements for supporting services
- Recommending solutions and validating them with stakeholders

## Scope beyond software

In IT, business analysis is applied most often to software development projects. But the manual warns explicitly against that restriction: confining business analysis to software development risks producing incomplete solutions. It applies equally to higher-level architectures, to services, and to the service value system as a whole.

The work is performed primarily by business analysts, though others contribute.

5.2.2

# Utility and warranty requirements

Business requirements fall into two families. The distinction mirrors utility and warranty from chapter 2.

**DEFINITION — UTILITY REQUIREMENTS**

Functional requirements that the customer has defined and that are unique to a specific product. What the service must do.

**DEFINITION — WARRANTY REQUIREMENTS**

Typically non-functional requirements captured from key stakeholders and other practices. Organisations should maintain a library of pre-defined warranty acceptance criteria for reuse.

### The analysis trap the manual warns against

- Do not analyse without any intention of acting on the analysis
- Do not analyse so deeply, or for so long, that a timely solution becomes impossible
- Do not try to solve every problem with one massive initiative that delivers too late to be useful

5.2.2

# What business analysis needs to succeed

**A** **Complete access to information**

Analysts need process flows, procedures, work instructions, policies and process metrics — and interviews not only with the process owner but with everyone who participates in each part of it.

**B** **A recognised skill set**

Critical thinking and evaluation, plus listening, communication and facilitation, the ability to document processes and use cases, and competence in data analysis and modelling.

**C** **Partner relationships across boundaries**

Where the system crosses organisational boundaries, the units involved must act as partners. Where compromises are needed, a collaborative relationship produces a solution that delivers value to all parties.

**D** **Interfaces with other practices**

Strategy management sets high-level direction; relationship management secures participation; service design, software development, and measurement and reporting all interact with the analysis.

5.2.3

# Capacity and performance management

**KEY MESSAGE**

The purpose is to ensure that services achieve the performance that has been agreed and expected, satisfying both current and future demand, and doing so cost-effectively.

**DEFINITION — PERFORMANCE**

A measure of what is achieved or delivered by a system, person, team, practice or service.

### Performance rests on capacity

Service capacity is the maximum throughput a configuration item or service can deliver. Specific metrics depend on the technology and the business nature of the service.

### Analysis

- Research and monitoring of current service performance
- Capacity and performance modelling

### Planning

- Capacity requirements analysis
- Demand forecasting and resource planning
- Performance improvement planning

5.2.3

# Capacity and performance across the value chain

### Plan

Supports tactical and operational planning with data on actual demand and performance, plus modelling and forecasting tools.

### Improve

Improvements are identified and driven by the performance information this practice generates.

### Engage

Customer and user expectations are managed using information about capacity constraints and capabilities.

### Design and transition

Essential to product and service design — ensures new services are designed for optimum performance, capacity and scalability.

### Obtain/build

Ensures components and services being built or obtained meet the organisation's performance needs.

### Deliver and support

Services are supported and tested against performance and capacity targets, metrics and reporting tools.

GH

Applied locally: student portal load at UPSA is not uniform. Registration windows and results release produce demand spikes that ordinary term-time capacity planning will miss entirely.

5.2.4

# Change control

KEY MESSAGE

The purpose is to maximise the number of successful service and product changes, by ensuring risks are properly assessed, authorising changes to proceed, and managing the change schedule.

DEFINITION — CHANGE

The addition, modification or removal of anything that could have a direct or indirect effect on services.

## Scope is set by the organisation

Typically all IT infrastructure, applications, documentation, processes, supplier relationships — and anything else that might affect a product or service directly or indirectly.

## The balance the practice must strike

Beneficial changes deliver additional value; the same changes can harm customers and users. Every change should therefore be assessed by people who genuinely understand both the risks and the expected benefits, and must be authorised before deployment. Crucially, that assessment must not introduce unnecessary delay.

5.2.4

# Three types of change, managed three different ways

## Standard changes

Low-risk, pre-authorised, well understood and fully documented. Implemented without additional authorisation, often initiated as service requests. The risk assessment is done once, when the procedure is created or modified — not on every repetition.

## Normal changes

Scheduled, assessed and authorised following a process. Change models determine the roles involved. Low-risk normal changes go to someone who can decide rapidly; very major ones may reach the management board. Triggered by a change request, which CI/CD pipelines increasingly automate.

## Emergency changes

Must be implemented as fast as possible — to resolve an incident or apply a security patch. Not normally in the change schedule; assessment and authorisation are expedited. Some documentation may be deferred, and a separate change authority of senior managers often applies.

*Examination tip: the distinguishing feature of a standard change is pre-authorisation, not low risk alone.*

5.2.4

# Change authority and the change schedule

DEFINITION — CHANGE AUTHORITY

The person or group who authorises a change. Assigning the correct authority to each type of change is what makes change control both efficient and effective.

## In high-velocity organisations

Decentralising change approval is common practice. Peer review then becomes one of the strongest predictors of high performance.

## What the change schedule does

- Helps plan changes and assign resources
- Assists communication and avoids conflicting changes
- After deployment, informs incident and problem management
- Feeds improvement planning

## Change control is not organisational change management

Organisational change management handles the people side of change, so that improvements and transformation initiatives actually take hold. Change control is focused on changes to products and services. Confusing the two is a frequent examination error.

PART TWO

# Operating the service: incidents, assets, events, problems and releases

- 5.2.5 Incident management
- 5.2.6 IT asset management
- 5.2.7 Monitoring and event management
- 5.2.8 Problem management
- 5.2.9 Release management

5.2.5

# Incident management

KEY MESSAGE

The purpose is to minimise the negative impact of incidents by restoring normal service operation as quickly as possible.

DEFINITION — INCIDENT

An unplanned interruption to a service, or a reduction in the quality of a service.

## Why it matters disproportionately

Incident management shapes customer and user satisfaction, and how the service provider itself is perceived, more visibly than almost any other practice.

## Non-negotiables

- Every incident is logged and managed so it is resolved within the time the customer and user expect
- Target resolution times are agreed, documented and communicated, so expectations stay realistic
- Incidents are prioritised using an agreed classification, so the highest business impact is resolved first
- Separate processes usually exist for major incidents and for information security incidents

5.2.5

# Who resolves an incident depends on its complexity

### 1 The user, via self-help

Use of specific self-help records should be captured for measurement and improvement.

### 2 The service desk

A substantial share of incidents is resolved at first contact.

### 3 A support team

More complex incidents are escalated, with routing usually based on incident category.

### 4 Suppliers or partners

Escalation to third parties who support their own products and services.

### 5 A temporary swarm team

For the most complex and all major incidents — drawing in provider, suppliers and users together.

### 6 Disaster recovery plans

In extreme cases, invoked to resolve the incident. Covered by service continuity management, 5.2.12.

GH All of these groups must understand the incident management process and how their contribution manages the value, outcomes, costs and risks of the service.

5.2.5

# Swarming, incident records and supplier support

DEFINITION — SWARMING

A technique in which many different stakeholders work together initially, until it becomes clear which of them is best placed to continue and which can move on to other tasks.

## Incident records and tooling

- Records stored in a suitable tool, linked to related CIs, changes, problems and known errors
- Modern ITSM tools match incidents automatically to prior incidents, problems or known errors
- Some now generate recommendations for handling future incidents
- Good collaboration tooling matters as much as good workflow tooling

## Quality of updates

Updates should cover symptoms, business impact, configuration items affected, actions completed and actions planned — each timestamped and attributed, so everyone involved or interested stays informed.

## Third parties in the chain

Third-party components require support agreements that align supplier obligations with the commitments the provider has made to its customers. Routine management of that part of the supplier contract often sits inside incident management. A supplier can even act as the service desk itself, logging and escalating on the provider's behalf.

5.2.6

# IT asset management

KEY MESSAGE

The purpose is to plan and manage the full lifecycle of all IT assets, so the organisation can maximise value, control costs, manage risks, support decisions about purchase, reuse, retirement and disposal, and meet regulatory and contractual requirements.

DEFINITION — IT ASSET

Any financially valuable component that can contribute to the delivery of an IT product or service.

DEFINITION — ASSET REGISTER

The accurate inventory information the practice depends on. Best populated by the processes that change asset status, not by periodic audit alone.

## Scope

All software, hardware, networking, cloud services and client devices. Sometimes non-IT assets such as buildings or information, where these carry financial value and are required to deliver an IT service. Increasingly it also includes operational technology, including Internet of Things devices that were never traditionally regarded as IT assets but now carry embedded computing and network connectivity.

5.2.6

# Asset management, ITAM and SAM

Three nested scopes. Students routinely use the terms interchangeably; the examination does not.

**1 Asset management**

A well-established practice covering the acquisition, operation, care and disposal of organisational assets, particularly critical infrastructure.

**2 IT asset management (ITAM)**

A sub-practice of asset management, aimed specifically at managing the lifecycles and total costs of IT equipment and infrastructure.

**3 Software asset management (SAM)**

An aspect of ITAM, aimed at the acquisition, development, release, deployment, maintenance and eventual retirement of software assets.

*Software runs on hardware, so software and hardware assets should be managed together to keep licences properly controlled — and cloud-based assets belong in the same regime.*

5.2.6

# Requirements differ sharply by asset type

### H Hardware assets

Must be labelled for identification. Location must be known, and assets protected from theft, damage and data leakage. Reuse or decommissioning may need special handling — disk erasure or shredding, depending on security requirements. Regulatory duties such as electrical waste directives may apply.

### S Software assets

Must be protected from unlawful copying that would create unlicensed use. Licence terms must be adhered to and reuse kept within the contract. Verified proof of purchase and entitlement must be retained. Licences are easily lost at decommissioning, so the process must recover and re-pool them.

### C Cloud-based assets

Must be assigned to specific products or groups so costs can be tracked. Funding must permit new instances when needed and removal when not — without uncontrolled cost. Contractual arrangements are handled as strictly as software licences.

### D Client assets

Must be assigned to named individuals who take responsibility for their care. Processes are needed for lost or stolen devices, together with tooling to erase sensitive data remotely or otherwise prevent its loss with the device.

GH Cost control warning: individual cloud usage looks cheap, but small amounts accumulate. Without ITAM discipline organisations consume far more than planned and receive a correspondingly large bill.

5.2.6

# Asset register, CMS federation and organisation

## Federating with the CMS

The asset register holds information about assets, their costs and related contracts, so it is often combined or federated with the configuration management system. Where the two stay separate, assets must be mappable between them — usually through a standard naming convention. Federation with financial and supplier systems may also be necessary.

## Centralised or distributed?

Some organisations run a centralised ITAM team, which may also own configuration management. Others make each technical team responsible for the assets it supports — storage team for storage, network team for network. Context and culture decide. But some central roles help ensure data quality and build expertise in licensing and inventory systems.

## Typical activities

- Define, populate and maintain the asset register — structure, content, and storage facilities for assets and related media
- Control the asset lifecycle with other practices — upgrading obsolete software, onboarding staff with a laptop and phone — and record every status, location, characteristic and assignment change
- Provide current and historical data, reports and support about IT assets to other practices
- Audit assets, media and conformity with regulations and licence terms, and drive corrective and preventive improvement

5.2.7

# Monitoring and event management

**KEY MESSAGE**

The purpose is to observe services and service components systematically, and to record and report selected changes of state identified as events. The practice identifies and prioritises infrastructure, service, business process and information security events, and establishes the appropriate response — including responses to conditions that could lead to faults or incidents.

**DEFINITION — EVENT**

Any change of state that has significance for the management of a service or other configuration item. Events are typically recognised through notifications created by an IT service, a CI, or a monitoring tool.

## The monitoring half

- Systematic observation of services and underpinning CIs
- Detects conditions of potential significance
- Should be highly automated; active or passive

## The event management half

- Records and manages state changes defined as events
- Determines significance and initiates the control action
- Not all monitoring yields an event

5.2.7

# Three classes of event, three different responses

## Informational

No action required at the time of identification. Value comes later: analysing the accumulated data may reveal desirable proactive steps that benefit the service.

## Warning

Allows action to be taken before any negative impact is actually experienced by the business. This is the pre-emptive band, and the reason monitoring earns its cost.

## Exception

A breach of an established norm has been identified — for example, against a service level agreement. Action is required even though business impact may not yet have been experienced.

### Key activities

- Decide what should be monitored and set the monitoring strategy
- Implement and maintain monitoring — native features plus purpose-built tools
- Establish thresholds and criteria defining each event type
- Set policies for handling each type, then automate them

### Active versus passive monitoring

In active monitoring, tools poll key configuration items and generate alerts when an exception condition appears. In passive monitoring, the configuration item itself generates the operational alert. Automated correlation of events matters as much as detection: volumes are enormous, and without policies on how to limit, filter and use the data it carries no value.

5.2.8

# Problem management

**KEY MESSAGE**

The purpose is to reduce the likelihood and impact of incidents, by identifying actual and potential causes of incidents and managing workarounds and known errors.

**DEFINITION — PROBLEM**

A cause, or a potential cause, of one or more incidents.

**DEFINITION — KNOWN ERROR**

A problem that has been analysed but has not been resolved.

**DEFINITION — WORKAROUND**

A solution reducing or eliminating the impact of an incident or problem for which full resolution is not yet available.

## Incidents

- Have an impact on users or business processes
- Must be resolved so normal business activity resumes
- The clock is the customer's

## Problems

- Are the causes of incidents
- Require investigation and analysis to identify causes
- Produce workarounds and longer-term resolutions, reducing future incidents

5.2.8

# The three phases of problem management

1

### Problem identification

Identify and log problems: trend analysis of incident records, duplicate and recurring issues spotted by users, desk and technical staff, recurrence risk noted during major incidents, plus information from suppliers, partners, developers and test teams.

2

### Problem control

Analyse problems and document workarounds and known errors. Problems are prioritised by risk and managed as risks, on impact and probability. Not every problem must be analysed — progress on the highest-priority ones beats investigating every minor one.

3

### Error control

Manage known errors where initial analysis is complete and faulty components identified. Identify potential permanent solutions, which may trigger a change request — but only where justified on cost, risk and benefit.

## Analyse across all four dimensions

Incidents typically have many interrelated causes. Problem control must consider every contributory cause, including those that lengthened duration or worsened impact. An incident caused by inaccurate documentation may require correcting the document and training support staff, suppliers and users.

## Living with workarounds

An effective workaround can become the permanent answer where resolution is not viable or cost-effective. The problem then remains a known error, and the documented workaround is applied when related incidents occur. Every workaround must define the symptoms it applies to — and its effectiveness should be re-evaluated each time it is used.

5.2.8

# Where problem management touches other practices

**A** **Risk management**

Problem management can be organised as a specific case of risk management — identifying, assessing and controlling risks across all four dimensions. Adopting risk tools and techniques is useful.

**B** **Change control**

Implementing a resolution usually sits outside its scope. It initiates resolution via change control and joins the post-implementation review, but approving and implementing changes is not its responsibility.

**C** **Knowledge management**

Output includes documentation of workarounds and known errors; the practice also draws on the knowledge management system to investigate, diagnose and resolve.

**D** **Continual improvement**

Improvement opportunities across all four dimensions may enter the continual improvement register, prioritised with continual improvement techniques and sometimes managed in a product backlog.

**GH** Complement or conflict: identifying a cause may resolve an incident faster — or investigating a cause may delay the actions needed to restore service. The two must be designed to work together.

5.2.9

# Release management

**KEY MESSAGE**

The purpose is to make new and changed services and features available for use.

**DEFINITION — RELEASE**

A version of a service or other configuration item, or a collection of configuration items, that is made available for use.

## What a release contains

Infrastructure and application components working together, plus documentation, training for users or IT staff, updated processes or tools — developed in house or procured and integrated.

## Scale and planning

- From one minor changed feature to a completely new service
- A release plan specifies the exact combination of components and the timing
- A release schedule documents timing, negotiated and agreed with customers
- A post-implementation review enables learning and confirms customer satisfaction

## Staged releases

Release is often staged: a pilot goes to a small group to confirm everything works before wider groups receive it. Sometimes staging is impossible — where a major restructuring of shared underlying data is required, everyone must receive the release simultaneously.

5.2.9

# Release in waterfall and in Agile/DevOps environments

## Traditional / waterfall

- Almost all release work happens before deployment
- Plans specify exactly which components deploy in the release
- Deployment then makes the new functionality available
- Release management and deployment may be combined into a single process

## Agile / DevOps

- Significant release activity happens after deployment
- Software and infrastructure deploy in many small increments
- Release activity enables the new functionality later — often as a very small change
- Typically integrated with the continuous integration and delivery toolchain

### Blue/green releases

Two mirrored production environments. Users are switched to the environment carrying the new functionality by network tools that connect them to the correct one.

### Feature flags

Specific features are released to individual users or groups in a controlled way. The functionality is deployed to production without being released; a user configuration setting then switches it on for chosen users.

PART THREE

# Describing, configuring, protecting and designing the service

- 5.2.10 Service catalogue management
- 5.2.11 Service configuration management
- 5.2.12 Service continuity management
- 5.2.13 Service design

5.2.10

# Service catalogue management

KEY MESSAGE

The purpose is to provide a single source of consistent information on all services and service offerings, and to make sure it reaches the relevant audience.

## Catalogue versus portfolio

The catalogue lists services currently available. It is a subset of the fuller list tracked in the provider's service portfolio. The catalogue can be a document, an online portal, or a tool.

## Ongoing activities

Publishing, editing and maintaining service and product descriptions and their offerings. It shows what is available, and on what terms. Supported by roles such as the service owner as services are introduced, changed or retired.

### 1 User views

Information on service offerings that can be requested, plus provisioning details.

### 2 Customer views

Service level, financial and service performance data.

### 3 IT to IT customer views

Technical, security and process information for use in service delivery.

5.2.10

# From static list to engagement tool

## The adoption problem

For the catalogue to be seen as useful by the customer organisation it must do more than publish information statically. Unless it enables customer engagement — supporting discussion of standard and non-standard offerings, or automating request and order fulfilment — its chances of ongoing adoption as a meaningful resource are minimal.

DEFINITION — REQUEST CATALOGUE

A view of the service catalogue providing details on service requests for existing and new services, made available to the user.

## Why organisations gravitate to it

Because the request catalogue focuses on the consumable, orderable elements of service offerings — the part users can act on directly.

GH

Applied locally: an intranet page listing IT services is not a service catalogue in this sense. Until a lecturer can request a projector replacement or a lab account from it, it remains a notice board.

Value chain: involved in all six activities, most visibly engage — enabling and partly automating relationship management, request management and the service desk.

5.2.11

# Service configuration management

**KEY MESSAGE**

The purpose is to ensure that accurate, reliable information about the configuration of services and the CIs supporting them is available when and where it is needed — including how CIs are configured and the relationships between them.

**DEFINITION — CONFIGURATION ITEM (CI)**

Any component that needs to be managed in order to deliver an IT service.

**DEFINITION — CONFIGURATION MANAGEMENT SYSTEM (CMS)**

A set of tools, data and information used to support service configuration management.

## What is collected, and why relationships matter more than counts

CIs typically include hardware, software, networks, buildings, people, suppliers and documentation — and services themselves are treated as CIs. The practice explains how the many CIs contributing to a service work together: how they interact, relate and depend on each other to create value, including dependencies between services. That high-level view is often called a service map or service model, and forms part of the service architecture.

5.2.11

# Balancing the cost of configuration data against its value

## The discipline the manual insists on

Maintaining large amounts of detailed information about every component and its relationships can be costly and may deliver very little value. Requirements must be based on the organisation's goals and on how configuration management actually contributes to value creation.

## Plan by asking four questions

- Who needs the configuration information?
- How will it be used?
- What is the best way for them to obtain it?
- Who can maintain and update it?

## CMDB, federation and sensitivity

- Information may sit in one CMDB, but is more commonly distributed across sources
- Links between configuration records must be maintained either way
- Some organisations federate CMDBs for an integrated view
- Some information is sensitive — useful to an attacker, or personal data about users

## Processes typically needed

- Identify new CIs and add them to the CMS
- Update configuration data when changes are deployed
- Verify that configuration records are correct
- Audit applications and infrastructure to find anything undocumented

5.2.12

# Service continuity management

KEY MESSAGE

The purpose is to ensure that the availability and performance of a service are maintained at sufficient levels in the event of a disaster. The practice provides a framework for organisational resilience, producing a response that safeguards key stakeholders and the organisation's reputation, brand and value-creating activities.

## Trigger threshold

It engages when a service disruption or organisational risk exceeds what normal response and recovery practices — including major incident management — can absorb. An event of that magnitude is termed a disaster.

## Defining a disaster in advance

Each organisation must define what counts as a disaster in its own context, at both organisational and per-service level, using a business impact analysis — and it must do so before a trigger event, not during one.

## Disaster sources the manual lists

Supply chain failure • Terrorism • Weather • Cyber attack • Health emergency • Political or economic event • Technology failure • Public crisis
Organisational impact ranges from lost income and damaged reputation to breach of law and safety regulations, and risk to personal safety.

5.2.12

# The four examinable continuity terms

## RTO — Recovery time objective

The maximum acceptable period following a disruption that can elapse before the absence of business functionality severely impacts the organisation — the maximum agreed time within which a product or activity must be resumed, or resources recovered.

## RPO — Recovery point objective

The point to which information used by an activity must be restored, so the activity can operate on resumption. In practice, how much data loss is tolerable.

## DRP — Disaster recovery plans

A set of clearly defined plans for how the organisation will recover from a disaster and return to a pre-disaster condition, considering all four dimensions of service management.

## BIA — Business impact analysis

Identifies vital business functions and their dependencies — suppliers, people, other processes, IT services. It defines recovery requirements: RTOs, RPOs and minimum target service levels for each IT service.

GH Distinguish carefully: RTO is about time to restore the service; RPO is about how far back the data must be recoverable. Examinations test the pair together.

5.2.12

# Service continuity or major incident? Draw the line first

## Service continuity management

- Focuses on events the business considers significant enough to be a disaster
- Invokes pre-defined disaster recovery plans
- Concerned with organisational survival and resilience
- Must be thought through, designed for flexibility and tested regularly

## Incident and major incident management

- Handles less significant events within normal response capability
- Restores normal service operation as quickly as possible
- Concerned with day-to-day service quality
- Escalates upward when thresholds are breached

### The distinction must be documented before the event

The manual is explicit: the distinction between disasters, major incidents and incidents needs to be pre-defined, agreed and documented, with clear thresholds and triggers for calling the next tier of response and recovery into action — without unnecessary delay and risk.

High availability itself comes from a combination: business planning, technical architecture resilience, availability planning, proactive risk and information security management, and incident and problem management working together.

5.2.13

# Service design

KEY MESSAGE

The purpose is to design products and services that are fit for purpose, fit for use, and deliverable by the organisation and its ecosystem. This includes planning and organising people, partners and suppliers, information, communication, technology and practices — and the interaction between the organisation and its customers.

## What happens without it

- Products and services become unduly expensive to run and prone to failure
- Resources are wasted; the service is neither customer-centred nor holistically designed
- No improvement programme is likely to achieve what proper design would have achieved first time
- Cost-effective services delivering what customers expect become extremely hard to achieve

## Design decisions never sit in isolation

- Impact on other products and services
- All relevant parties, including customers and suppliers
- The existing architectures and the required technology
- The service management practices themselves
- The necessary measurements and metrics
- And retirement — an unplanned withdrawal harms customers avoidably

5.2.13.1

# Design thinking

A practical, human-centred approach that accelerates innovation — complementary to Lean and Agile, drawing on logic, imagination, intuition and systems thinking.

1

### Inspiration and empathy

Direct observation of people and how they work with products and services, and how they might interact differently.

2

### Ideation

Divergent thinking generates many unique ideas; convergent thinking narrows them to a preferred solution.

3

### Prototyping

Ideas tested early, iterated and refined. Prototypes gather feedback and expose strengths and weaknesses fast.

4

### Implementation

Concepts brought to life, coordinated with all relevant practices. Agile can deliver the solution iteratively.

5

### Evaluation

With project and release management, measures actual performance against acceptance criteria and finds improvement opportunities.

### Why multi-disciplinary teams

Design thinking balances the perspectives of customers, technology, the organisation, partners and suppliers. That makes it highly integrative, well aligned to the service value system, and a key enabler of digital transformation.

5.2.13.2

# Customer experience, user experience and Lean UX

## Customer experience (CX)

- Manages every aspect of the complete experience
- Covers time, quality, cost, reliability and effectiveness
- The whole journey from demand to value realisation

## User experience (UX)

- Looks specifically at ease of use of the product or service
- Concerns how the customer interacts with it
- Narrower, more immediate, more testable

### Lean UX

A mindset, a culture and a process embracing Lean–Agile methods. It implements functionality in minimum viable increments and judges success by measuring results against an outcome hypothesis. The core objective is to obtain feedback as early as possible so that quick decisions can be made.

### Typical Lean UX questions

- Who are the customers of this product or service, and what will it be used for?
- When is it used, and under what circumstances?
- What will be the most important functionality?
- What are the biggest risks?
- Assumptions are then prioritised by the risk they represent

PART FOUR

# Meeting the user: desk, service levels, requests and testing

- 5.2.14 Service desk
- 5.2.15 Service level management
- 5.2.16 Service request management
- 5.2.17 Service validation and testing

5.2.14

# Service desk

KEY MESSAGE

The purpose is to capture demand for incident resolution and service requests. It should also be the entry point and single point of contact between the service provider and all of its users.

## The model varies; the function does not

Delivery may run from a physical team working shifts, to a distributed mix of people connected virtually, to automated technology and bots. The function and the value remain the same regardless of the model.

## The shift in what desks do

With increased automation and the gradual removal of technical debt, the focus has moved to supporting people and business rather than technical issues alone. Desks increasingly arrange, explain and coordinate rather than only fix broken technology.

## Two points the manual stresses

- However efficient the desk, some issues will always need escalation and underpinning support — support and development teams must collaborate closely to present a joined-up approach
- The desk should be the empathetic and informed link between provider and users, adding value by understanding and acting on the business context of an action, not merely logging it

5.2.14

# Channels, technology and skills

## Access channels

- Phone, including IVR, conference calls and voice recognition
- Service portals and mobile applications, backed by catalogues and knowledge bases
- Live chat and chatbots
- Email for logging, updating and follow-up surveys
- Walk-in desks, increasingly common in higher education
- Text and social media messaging; public and corporate forums for peer support

## Supporting technology for a centralised desk

- Intelligent telephony — computer-telephony integration, IVR, automatic call distribution
- Workflow systems for routing and escalation
- Workforce management and resource planning systems
- A knowledge base; call recording and quality control
- Remote access tools; dashboards and monitoring
- Configuration management systems

## Virtual desks

Agents work from multiple geographically dispersed locations. This requires more sophisticated technology, with more complex routing and escalation — solutions are often cloud-based.

## Skills that matter most

Empathy, incident analysis and prioritisation, effective communication and emotional intelligence. The key skill is understanding and diagnosing an incident in terms of business priority, then acting to resolve it with the skills, knowledge, people and processes available.

5.2.15

# Service level management

**KEY MESSAGE**

The purpose is to set clear, business-based targets for service levels, and to ensure that service delivery is properly assessed, monitored and managed against those targets.

**DEFINITION — SERVICE LEVEL**

One or more metrics that define expected or achieved service quality.

**DEFINITION — SERVICE LEVEL AGREEMENT (SLA)**

A documented agreement between a service provider and a customer identifying both the services required and the expected level of service.

## Service level management provides end-to-end visibility. To achieve it, the practice:

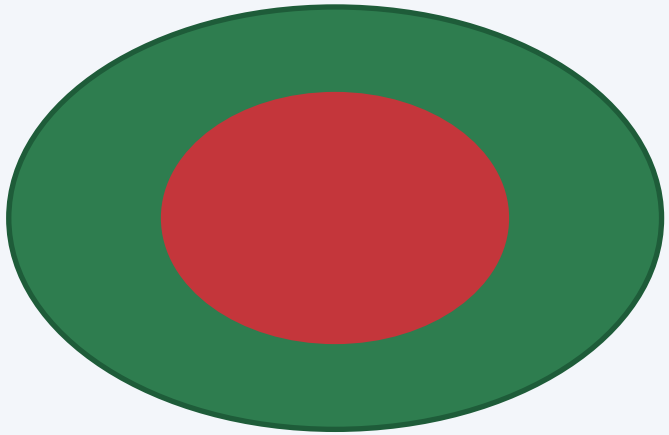
- Establishes a shared view of services and target service levels with customers
- Ensures defined levels are met, through collection, analysis, storage and reporting of relevant metrics
- Performs service reviews, confirming the current set of services still meets organisational and customer needs
- Captures and reports service issues, including performance against defined service levels

5.2.15

# The watermelon SLA effect

## Green outside, red inside

Traditional SLAs were built on individual activities: incident resolution times, system availability figures, volume counts. Without business context these are often meaningless. A system availability of 99.6 per cent looks impressive — but the acceptable 0.4 per cent of unavailability becomes intolerable if it falls during a commercial transaction, an operating theatre in use, or point-of-sale tills in use. Satisfaction will be low regardless of whether the SLA was met.



*SLA reports*

## Requirements for a successful SLA

- Related to a defined service in the catalogue, not a free-floating metric
- Related to defined outcomes, using balanced bundles of metrics
- A genuine agreement, discussed with partners, sponsors, users and customers
- Simply written, easy to understand and use for all parties

## The corrective

Service level management identifies metrics that truthfully reflect the customer's actual experience and satisfaction with the whole service. These vary across organisations, and the only way to learn what they are is to find out directly from customers.

5.2.15

# What service level management listens to

### 1 Customer engagement

Initial listening, discovery and information capture. Open questions work best: what differentiates a good day from a bad one, and what is the best measure of your success?

### 2 Customer feedback

Gathered formally and informally. Surveys range from immediate follow-up questions after an incident to reflective periodic surveys on overall experience — both are event-based.

### 3 Key business-related measures

Agreed on what the customer values: a bundle of SLA metrics, or a very specific activity such as a sales transaction, a project completion, or getting an ambulance to an accident within a stated time.

### 4 Operational and business metrics

Operational metrics are low-level indicators — availability, response and fix times, request processing times. Business metrics are any activity the customer values, such as passenger check-in.

5.2.16

# Service request management

KEY MESSAGE

The purpose is to support the agreed quality of a service by handling all pre-defined, user-initiated service requests in an effective and user-friendly manner.

DEFINITION — SERVICE REQUEST

A request from a user, or a user's authorised representative, that initiates a service action agreed as a normal part of service delivery.

### A service delivery action

Providing a report, replacing a toner cartridge

### Information

How to create a document, or what the office hours are

### Provision of a resource

A phone or laptop for a user, a virtual server for a development team

### Access to a resource

Access to a particular file or folder

### Feedback and complaints

Complaints about a new interface, or compliments to a support team

5.2.16

# Making service requests work at scale

### Why standardisation is possible here and not elsewhere

Because requests are pre-defined and pre-agreed as a normal part of service delivery, they can be formalised with a clear standard procedure for initiation, approval, fulfilment and management. Some have very simple workflows — a request for information. Others, such as setting up a new employee, are complex and draw on many teams and systems. Either way, the steps must be well known and proven so fulfilment times can be agreed and status communicated.

**1 Standardise and automate**

Standardise and automate requests and their fulfilment as far as possible.

**2 Streamline approvals**

Set policies on which requests need limited or no extra approval.

**3 Set expectations honestly**

Base fulfilment-time expectations on what can realistically be delivered.

**4 Improve continuously**

Find opportunities for faster fulfilment and more automation.

**5 Redirect misfiled work**

Redirect requests that are really incidents or changes.

**6 Reuse workflow models**

Few workflow models, reused for new requests, aid maintainability.

5.2.17

# Service validation and testing

**KEY MESSAGE**

The purpose is to ensure that new or changed products and services meet defined requirements. Service value is defined from customer input, business objectives and regulatory requirements, documented within design and transition, and used to establish measurable quality and performance indicators supporting assurance criteria and testing requirements.

## Service validation (5.2.17.1)

Establishes deployment and release acceptance criteria — the conditions for production readiness — which are then verified through testing. Criteria may be utility- or warranty-focused, derived from customer, regulatory, business, risk and security requirements.

## Test strategy (5.2.17.2)

Defines the overall approach to testing. It can apply to an environment, a platform, a set of services or one service. It applies equally to in-house and externally developed solutions, and must match the stakeholders' risk appetite.

GH

Applied locally: procurement of a vendor-built system does not transfer the testing obligation. The manual is explicit that externally developed solutions are tested on the same footing as in-house builds.

*Value chain: involved in every activity except plan — with the heaviest contribution to design and transition, and obtain/build.*

5.2.17.2

# Typical test types: utility and warranty

## Utility / functional tests

- Unit test — a test of a single system component
- System test — overall testing of the system, including software and platforms
- Integration test — testing a group of dependent software modules together
- Regression test — testing whether previously working functions were impacted

## Warranty / non-functional tests

- Performance and capacity test — speed and capacity under load
- Security test — vulnerability, policy compliance, penetration, denial of service risk
- Compliance test — legal and regulatory requirements have been met
- Operational test — backup, event monitoring, failover, recovery and reporting
- Warranty requirements test — documentation, training, support model, knowledge transfer
- User acceptance test — performed by users to approve a release

### The metrics that matter

Escaped defects, test coverage and service performance against SLAs are critical success measures — they are what allow customer experience to improve and risk to fall.

PART FIVE

# Contemporary issues in the field

- AIOps and machine learning in operations
- Cloud economics, FinOps and asset sprawl
- Site reliability engineering and platform engineering
- Cybersecurity, data protection and resilience in Ghana

CONTEMPORARY ISSUE 1

# AIOps: machine learning inside the operational practices

The manual already anticipates this — modern ITSM tools automatically match incidents to prior incidents, problems and known errors, and generate recommendations. The direction of travel since has been sharper.

### 1 Event noise reduction and correlation

Machine learning collapses thousands of correlated alerts into a handful of actionable signals — directly addressing the manual's warning that huge data volumes without filtering policies carry no value.

### 2 Anomaly detection replacing static thresholds

Rather than fixed warning and exception thresholds, models learn normal behaviour per service and per season. This changes how monitoring and event management sets its criteria.

### 3 Assisted diagnosis in problem management

Causal inference over incident and change records surfaces candidate causes. Human judgement still decides — the manual insists diagnosis relies on knowledge and experience, not procedure.

### 4 Conversational first line at the service desk

Large language models handle a growing share of requests and simple incidents. The practice question becomes where the empathetic, informed human link is still required.

CONTEMPORARY ISSUE 2

# Cloud economics, FinOps and the sprawl problem

### The manual's warning, restated

Each individual use of a cloud service may be relatively cheap, but spending in small amounts makes it easy to consume far more resource than planned — leaving the organisation with a correspondingly large bill. FinOps is the discipline that grew up around exactly this failure.

### 1 Showback and chargeback

Cloud assets assigned to specific products or groups so cost is visible to the team that generates it — the manual's own requirement, now operationalised.

### 2 Rightsizing and reservation

Continuous matching of provisioned capacity to observed demand, bringing capacity and performance management directly into the cost conversation.

### 3 Licence entitlement in the cloud

Bring-your-own-licence models make SAM harder, not easier. Entitlement records must survive migration or organisations pay twice.

### 4 SaaS discovery

Departments procure SaaS without IT involvement. The asset register cannot record what it never learns about, and audits become the only detection mechanism.

**GH** Applied locally: foreign-currency cloud billing exposes Ghanaian institutions to exchange-rate movement on top of consumption growth. Budgeting in cedis against a dollar-denominated bill is itself a capacity and financial management risk.

CONTEMPORARY ISSUE 3

# SRE, DevOps and platform engineering

## Where ITIL 4 already meets SRE

- The manual names site reliability engineers as owners of availability for specific products
- Anti-fragile design and MTRS minimisation are core SRE positions
- Decentralised change approval with peer review as a performance predictor is a DevOps finding
- Blue/green releases and feature flags are described directly in release management

## Where the tension remains

- Error budgets set a deliberate tolerance for failure; SLAs traditionally do not
- Toil reduction can conflict with the documentation expectations of change control
- You build it, you run it blurs the service desk's single point of contact
- Platform engineering creates internal developer platforms that behave as services with their own catalogues, SLAs and desks

### The synthesis that most organisations arrive at

Practices remain the shared vocabulary and the accountability structure; engineering culture supplies the speed. ITIL 4 was deliberately rewritten to permit this — which is why change control replaced change management, and why the manual repeatedly warns that assessment must not introduce unnecessary delay.

CONTEMPORARY ISSUE 4

# Security, data protection and resilience in the Ghanaian context

### A Data protection duties inside configuration management

The manual warns that configuration information may include personal data such as phone numbers and home addresses. Under Ghana's Data Protection Act, 2012 (Act 843), that makes the CMDB itself a processing activity requiring lawful basis and safeguards.

### B Cybersecurity as a continuity trigger

Cyber attack sits in the manual's list of disaster sources. The Cybersecurity Act, 2020 (Act 1038) and the incident reporting expectations that follow from it turn parts of incident and continuity management into statutory obligations for designated critical information infrastructure.

### C Power and connectivity as availability constraints

MTBF and MTRS calculations that assume stable grid supply understate real risk. Availability design in Ghana must treat power continuity and redundant connectivity as first-class components, not facilities issues.

### D Public-sector service catalogues and citizen expectation

Digital public services — tax, procurement, identity and payments platforms — are now consumed directly by citizens. Watermelon SLAs are politically costly when the user is a taxpayer rather than an internal colleague.

CONSOLIDATION

# Which practices concentrate where in the value chain

| | |
|---|---|
| **Involved in all six activities** | Availability management, business analysis, capacity and performance management, change control, release management, service catalogue management, service configuration management, service continuity management, service design |
| **All activities except plan** | Monitoring and event management, service desk, service request management, service validation and testing |
| **Concentrated in engage, and deliver and support** | Incident management |
| **Concentrated in improve, and deliver and support** | Problem management |
| **Concentrated in design and transition, and obtain/build** | IT asset management |
| **Concentrated in plan and engage** | Service level management |

*Read as typical contribution, not as a maturity or priority ranking.*

KEY TAKEAWAYS

# Six things to carry out of this session

### 1 Practices are resources, not procedures

Each bundles people, information, technology, partners and workflows across all four dimensions.

### 2 Restoration and diagnosis are different jobs

Incident management restores service. Problem management finds and removes causes. Designed together, they complement; conflated, they conflict.

### 3 Measurement must reflect experience

Percentage availability and single-system SLA targets can be met while customers are failed. Ask customers what success means.

### 4 Control must not become delay

Correct change authority, pre-authorised standard changes and decentralised approval with peer review are how control and speed coexist.

### 5 Know the threshold before the event

Incident, major incident and disaster must be distinguished, agreed and documented in advance, with clear triggers.

### 6 Design decisions are never isolated

Every change to a service touches other services, parties, architectures, practices and metrics — including its eventual retirement.

ASSESSMENT PREPARATION

# Review and discussion questions

### Recall — attempt without notes

- Define incident, problem, known error and workaround, and state how each relates to the others.
- Distinguish standard, normal and emergency changes by their authorisation route.
- State the difference between RTO and RPO.
- Which practices are involved in every value chain activity except plan?

### Application — 10 marks each

- A university portal is available 99.7 per cent of the year but fails during every registration window. Diagnose the failure using service level management.
- Design a request catalogue for a Ghanaian district assembly's IT unit, naming five request types and their workflows.

### Evaluation — seminar debate, 20 marks

- “In an organisation practising continuous deployment, formal change control adds cost without adding safety.” Argue for and against, using evidence from section 5.2.4, and reach a defended position.
- Assess whether the availability measurements in section 5.2.1 are adequate for a mobile money service operating in a market with intermittent power supply. Recommend and justify two additional measures.

REFERENCES

# Reading and acknowledgements

## Primary source

AXELOS (2019). ITIL® Foundation: ITIL 4 Edition. London: TSO. Chapter 5, sections 5.2 to 5.2.17.2 — ITIL management practices.

All definitions, purpose statements and value chain contributions in this deck are summarised from that chapter. ITIL® is a registered trade mark of AXELOS Limited.

## Recommended supplementary reading

- AXELOS, ITIL 4: Create, Deliver and Support (for practice depth)
- Beyer et al., Site Reliability Engineering (for the MTRS argument)
- Forsgren, Humble and Kim, Accelerate (for peer review as a performance predictor)
- Ghana Data Protection Act, 2012 (Act 843); Cybersecurity Act, 2020 (Act 1038)

## Preparation for the next session

Read section 5.1, the general management practices, and section 5.3, the technical management practices. Come prepared to argue which single practice from section 5.2 your own organisation performs worst, and what evidence supports that judgement.

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ITIL® 4 Foundation

# Technical Management Practices

## Chapter 5, Sections 5.3 – 5.3.3

Deployment management · Infrastructure and platform management · Software development and management

*Lecture slides prepared for undergraduate and postgraduate students in Information Systems*

ORIENTATION

# Session at a Glance

**3**

### Practices covered

Deployment, infrastructure and platform, software development and management

**4**

### Deployment approaches

Phased, continuous delivery, big bang and pull deployment

**3**

### Cloud service models

SaaS, PaaS and IaaS — plus four cloud deployment models

**6**

### Value chain activities

Each practice mapped onto the ITIL service value chain

## Why these three practices matter

Technical management practices are where service management meets the technology itself. They are adapted from technology management domains and repurposed for service management — they explain how components are built, hosted and moved into live use so that value can actually be co-created with customers.

LEARNING OUTCOMES

# Learning Objectives (1 of 2)

*By the end of this session, students should be able to:*

1

## Explain the purpose of the deployment management practice

State the key message and describe how it moves new or changed components into live environments.

2

## Distinguish deployment from release management and change control

Explain why they are closely related but remain separate practices, including the term ‘provisioning’.

3

## Compare the four deployment approaches

Phased deployment, continuous delivery, big bang deployment and pull deployment — and when each is appropriate.

4

## Describe the definitive media library and definitive hardware store

Explain why deployable components must be held in secure, controlled locations.

# Learning Objectives (2 of 2)

5

## Explain the purpose and scope of infrastructure and platform management

Including physical, virtual and cloud resources, and the facilities that house them.

6

## Differentiate the cloud service and cloud deployment models

SaaS, PaaS and IaaS; private, public, community and hybrid cloud.

7

## Analyse how cloud computing reshapes other ITIL practices

Service financial management, supplier management, capacity and performance, change control, incident and deployment management.

8

## Describe the software development and management practice and lifecycle

From ideation through design, develop, test, deploy and operate, to retirement.

9

## Map each practice onto the service value chain

Interpret the heat maps in Figures 5.37, 5.38 and 5.40 and justify the ratings.

STRUCTURE

# How This Session Is Organised

**5.3**

**Technical management practices**

The category and its place in ITIL 4

›

**5.3.1**

**Deployment management**

Approaches, DML/DHS, tooling, value chain

›

**5.3.2**

**Infrastructure and platform management**

Cloud models, practice impacts

›

**5.3.3**

**Software development and management**

Activities, lifecycle, value chain

**Then: Contemporary issues · Ghanaian application · Key terms · Self-assessment · Tutorial activity**

*The examinable content is the ITIL 4 text; the contemporary issues section extends it to current practice in 2026.*

CONTEXT

# Where 5.3 Sits: The 34 ITIL Management Practices

**14**

### General management practices

Adopted and adapted from general business management — e.g. strategy management, risk management, information security management, supplier management.

**17**

### Service management practices

Developed in service management and ITSM — e.g. incident management, change enablement, service desk, service level management.

**3**

### Technical management practices

Adapted from technology management domains for service management — deployment management, infrastructure and platform management, software development and management.

*A practice is a set of organisational resources designed for performing work or accomplishing an objective. This session covers the three technical management practices in full.*

# 5.3.1

# Deployment Management

Moving new or changed components into live environments — and into test and staging environments too.

**Purpose**

**Four approaches**

**DML & DHS**

**Value chain**

# Purpose of the Practice

**KEY MESSAGE**

The purpose of the deployment management practice is to move new or changed hardware, software, documentation, processes, or any other component to live environments. It may also be involved in deploying components to other environments for testing or staging.

## What is deployed?

Hardware, software, documentation, processes — or any other component. It is not a software-only practice.

## Where to?

Primarily live environments, but also test and staging environments.

## Why does it matter?

Nothing a service provider builds creates value until it reaches the environment where customers consume it.

# Deployment, Release and Change Control

## Deployment management

Moves new or changed components to live (and test/staging) environments.

***Technical act of moving and installing***

## Release management

Makes new or changed services and features available for use. Owns communication with users and customers.

***Business act of making available***

## Change control

Maximises successful changes by ensuring risks are properly assessed and changes authorised.

***Governance act of authorising***

### Closely related — but separate practices

A component can be deployed to the live environment long before it is released to users. Individual deployments are not generally of interest to users and customers until they are released — which is why communication around deployments belongs to release management.

# A Note on Terminology: ‘Provisioning’

In some organisations

**‘Provisioning’ = infrastructure**

**‘Deployment’ = software only**

The two words are used to separate the hardware/infrastructure activity from the software activity.

In ITIL 4

**‘Deployment’ means BOTH**

The term deployment is used to mean the movement of infrastructure and of software — the practice covers hardware, software, documentation and processes alike.

**Examination point**

Candidates frequently lose marks by treating deployment as a software-only activity. Remember: deploying a batch of laptops to a branch office, publishing an updated operating procedure, or racking a new server are all deployment management.

# Four Distinct Deployment Approaches

Many organisations use a combination of these approaches, depending on their specific services and requirements as well as the release sizes, types, and impact.

**01**

## Phased deployment

Part of the production environment at a time, repeated until complete

**02**

## Continuous delivery

Integrated, tested and deployed when needed, with frequent feedback loops

**03**

## Big bang deployment

All targets at the same time, where dependencies demand it

**04**

## Pull deployment

Controlled repository; users download to client devices when they choose

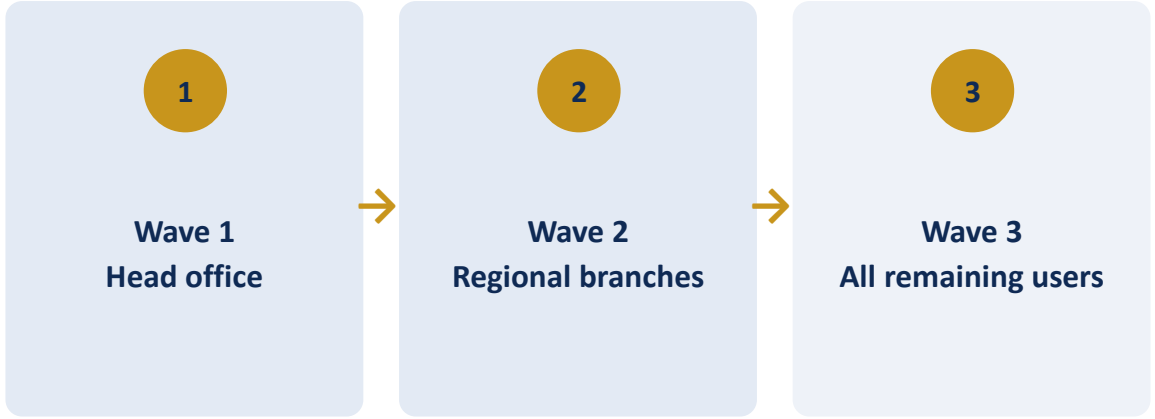
*Each approach is examined on the following slides, then compared side by side.*

5.3.1 APPROACH 1 OF 4

# Phased Deployment

**Definition**

The new or changed components are deployed to just part of the production environment at a time — for example to users in one office, or one country. The operation is repeated as many times as needed until the deployment is complete.




## Strengths

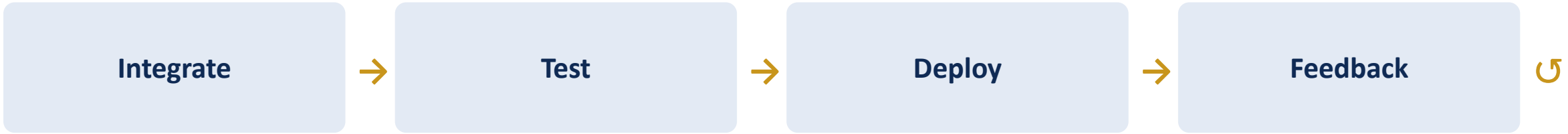
- Exposure is limited — problems affect one group only
- Lessons from earlier waves improve later ones
- Support effort is spread over time

## Trade-offs

- Two versions coexist — configuration must tolerate this
- Longer overall elapsed time to complete
- Requires accurate records of who has what

# Continuous Delivery

Components are integrated, tested, and deployed when they are needed, providing frequent opportunities for customer feedback loops.




*Repeats continuously — the loop, not the event, is the unit of work*

### Depends on automation

An automated toolchain for build, test and deployment is effectively a precondition.

### Small batch sizes

Each change is small, so failure is cheap to detect and cheap to reverse.

### Feedback is the product

Frequent customer feedback loops are the stated benefit — speed is a means, not the end.

5.3.1 APPROACH 3 OF 4

# Big Bang Deployment

**Definition**

New or changed components are deployed to all targets at the same time. This approach is sometimes needed when dependencies prevent the simultaneous use of both the old and new components.

**Worked example from the text**

**A database schema change that is not compatible with previous versions of some components.**

If old and new cannot read the same data structure, they cannot run side by side — so a phased rollout is simply not available.

## When it is the right choice

- Hard technical dependencies rule out coexistence
- Regulatory cut-over dates apply to everyone at once
- The change is small and thoroughly tested

## What it demands

- A rehearsed, tested rollback or remediation plan
- Support capacity concentrated on the cut-over window
- Full stakeholder communication through release management

# Pull Deployment

New or changed software is made available in a controlled repository, and users download the software to client devices when they choose.

### 1 Controlled repository

The organisation publishes an approved, verified package

### 2 User decision

The user chooses when to pull the update to the device

### 3 Service request management

Integration lets users request software only when it is needed

**Two stated benefits: users control the timing of updates, and licences or bandwidth are consumed only when software is genuinely needed.**

# The Four Approaches Compared

| Approach | Unit of deployment | Typical trigger | Chief risk | Best suited to |
|---|---|---|---|---|
| **Phased** | One part of the production environment at a time | Large user base, geographically spread | Two versions coexisting in live use | Branch networks, national rollouts |
| **Continuous delivery** | Each small change, whenever it is ready | A completed and tested increment | Weak automation lets defects through fast | Digital products, DevOps teams |
| **Big bang** | Every target simultaneously | Dependencies that block coexistence | Wide blast radius if it fails | Schema changes, regulatory cut-overs |
| **Pull** | Whatever the user chooses to download | A user decision or a service request | Estate drifts out of a known state | Client applications, mobile estates |

*Most organisations combine approaches — the choice depends on the service, the release size, the release type and its impact.*

# Choosing an Approach: Four Questions

### ? Can the old and new versions coexist?

If not, big bang may be unavoidable. If they can, phased or continuous delivery become available.

### ? How large and how risky is the release?

Release size, type and impact are the stated determinants. Big releases with wide impact argue for phasing.

### ? How mature is the automated toolchain?

Continuous delivery is only sustainable where integration, testing and deployment are automated end to end.

### ? Who should control the timing?

If the user should decide — for bandwidth, licence cost or workload reasons — pull deployment is the fit.

*No approach is inherently superior — examiners test judgement about fit, not preference.*

# Securing What Is To Be Deployed

Components available for deployment should be maintained in one or more secure locations, to ensure they are not modified before deployment.

## DML

### Definitive media library

The secure location for software and documentation — the authoritative copies from which deployments are made.

## DHS

### Definitive hardware store

The secure location for hardware components held ready for deployment or for replacement of failed items.

**The controlling idea: integrity**

What is deployed must be exactly what was built, tested and authorised. Secure, controlled storage is the mechanism that makes that claim verifiable — and it is why these libraries are normally integrated with service configuration management.

# Tools That Support Deployment

Tools that support deployment are many and varied. Three integration points matter most:

1

### Configuration management tools

Deployment tools are often integrated with configuration management tools, so the CMDB reflects what has actually been deployed.

2

### Audit and change management

Integrated tooling provides the evidence trail that a deployment was authorised and executed as approved.

3

### Service portal and request management

Most organisations have tools for deploying client software; these may be integrated with a service portal to support a request management practice.

Tooling does not remove the need for a process. Most organisations have a documented deployment process, supported by standard tools and detailed procedures, so that software is deployed in a consistent way.

# Deployment When Infrastructure Is a Service

### Infrastructure as code

Where infrastructure is provided as a service, deployment of new or changed servers, storage or networking is typically managed by the organisation — often treating the infrastructure as code, so that deployment can be automated.

### The complication

Some deployments may be under the control of the supplier — firmware updates, or operating system patches where the supplier also provides the operating system.

## The obligation that survives outsourcing

The IT organisation must ensure that it knows what deployments are planned, and which have happened, in order to maintain a controlled environment. Control of the activity may be delegated; awareness of it cannot be.

# Third Parties, Multiple Suppliers, Multiple Processes

## Application development as a service

Deployment may be carried out by the external application developer, by the in-house IT department, or by a service integrator. The organisation must still be aware of all deployments so a controlled environment is maintained.

## Environments with multiple suppliers

It is important to understand the scope and boundaries of each organisation's deployment activities, and how these will interact — gaps and overlaps between suppliers are where control is lost.

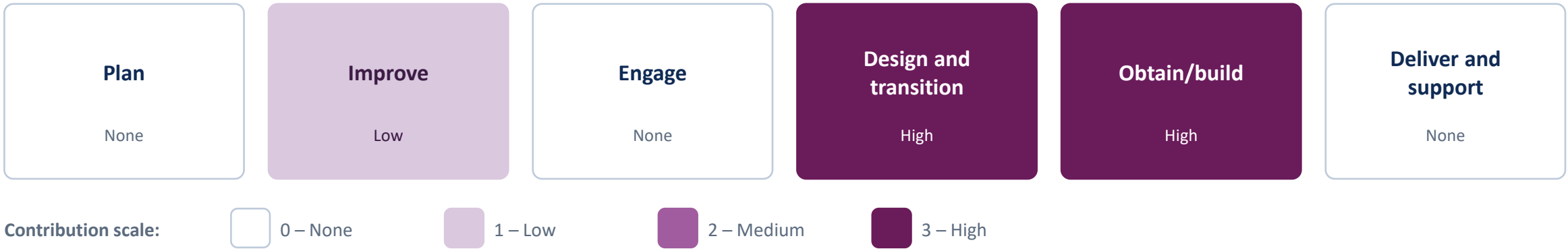
### Different environments, different processes

It is common to have different processes for different environments. For example, there may be one process for the deployment of client application software, and a completely different process for the deployment of server operating system patches — the risk profiles, approvals and rollback options are simply not the same.

FIGURE 5.37

# Deployment Management and the Service Value Chain

The practice is applied mainly to the design and transition and obtain/build activities, but also to improve.




## Improve

Some improvements require components to be deployed before they can be delivered; these are planned and managed in the same way as any other deployment.

## Design and transition

Deployment management moves new and changed components to live environments, so it is a vital element of this activity.

## Obtain/build

Changes can be deployed incrementally as part of this activity — especially common in DevOps environments using a complete automated toolchain for continuous integration, delivery and deployment.

THE ITIL STORY

# Axle Car Hire: Deployment Management

**M** **Marco**

*Before changes are deployed to the booking app, they are released to a test environment. After thorough testing, the changes are made available to users and customers.*

**R** **Radhika**

*The same logic applies to non-digital services. Two new hybrid models were introduced for hire in bigger cities: a promotional service offering was created, marketing materials updated, technicians trained, and everything deployed in advance — including the vehicles.*

**S** **Su**

*By the time the launch date arrived, the team was ready. The cars were made available to hire that very day.*

**H** **Henri**

*Partnering with the manufacturer meant a successful, well-prepared launch that created a buzz with customers — and with the manufacturer's own customers.*

*Note that the deployment happened before the official launch — and with the manufacturer's permission.*

ANALYSIS

# Three Lessons from the Axle Case

1

## Deployment is not release

The vehicles, materials, trained technicians and service offering were all in place before customers could hire the cars. Deployment preceded release by design.

2

## The practice is not software-only

Cars, marketing materials and staff training were deployed. ITIL deliberately defines the component broadly — hardware, software, documentation, processes.

3

## Deployment reaches across the value network

Coordination with the manufacturer determined the timing. In multi-party arrangements, deployment scope and boundaries must be agreed, not assumed.

APPLICATION

# Deployment Management in the Ghanaian Context

### Phased across a branch network

A rural and community bank deploying a new core-banking client to branches in Accra first, then regional capitals, then rural agencies — limiting exposure while connectivity varies by location.

### Big bang where the regulator sets the date

A regulatory reporting change with a single national effective date leaves no room for coexistence: all institutions cut over together.

### Pull deployment where bandwidth is costly

Publishing a large update to a controlled repository lets staff on metered mobile data pull it when they reach a campus or office network.

### Supplier-controlled deployments

Where a bank or university buys a hosted platform from an external provider, patches may be deployed by that provider — the institution must still know what changed, and when.

*Illustrative teaching examples — discuss which approach your own organisation actually uses, and why.*

# 5.3.2

# Infrastructure and Platform Management

Overseeing the infrastructure and platforms used by an organisation — including the technology of external service providers.

**Purpose & scope**

**Cloud service models**

**Deployment models**

**Practice impacts**

# Purpose of the Practice

KEY MESSAGE

The purpose of the infrastructure and platform management practice is to oversee the infrastructure and platforms used by an organization. When carried out properly, this practice enables the monitoring of technology solutions available to the organization, including the technology of external service providers.

## Oversee

The practice is a stewardship role over the whole technology estate, not merely an operations rota.

## Monitor

It enables monitoring of the technology solutions available to the organisation.

## Including external providers

Supplier-hosted technology is inside the scope. Ownership does not define the boundary; use does.

# What Counts as IT Infrastructure?

The physical and/or virtual technology resources that provide the environments needed to deliver IT services.

- **Servers**
- **Storage**
- **Networks**
- **Client hardware**
- **Middleware**
- **Operating systems**

**Also included**

Any configuration item a customer uses to access the service or consume a product — and the buildings and facilities the organisation uses to run its IT infrastructure.

**Who manages it?**

IT infrastructure may be managed by the service provider, or by an external supplier as dedicated, shared or cloud services. All three arrangements fall within the practice.

# Providing Technology That Creates Value

The practice includes the provision of technology needed to support activities that create value for the organisation and its stakeholders — which includes being ready to adopt new technologies.

- **Machine learning**
- **Chatbots**
- **Artificial intelligence**
- **Mobile device management**
- **Enterprise mobility management**

*Technologies named in the ITIL 4 text as examples of readiness to adopt*

'Readiness' is the operative word: the practice is judged not only on running today's estate well, but on whether the organisation is positioned to absorb the next technology without disruption.

# Every Organisation Needs Its Own Strategy

## No universal blueprint

Every single organisation must develop its own strategy to achieve the intended outcome with any type of infrastructure or platform. Copying another organisation's architecture does not transfer its context, constraints or goals.

## A cloud management system

Each organisation should design its own cloud management system to orchestrate all the interrelated components of infrastructure and platform with its business goals, the intended service quality and operational efficiency.

**Business goals**

**Intended service quality**

**Operational efficiency**

*The three things a cloud management system must be orchestrated against*

**Exam tip: the text says design your own — not adopt a standard one.**

# Cloud Service Models: SaaS, PaaS, IaaS

## SaaS

**Software as a service**

The consumer can use the applications running in the cloud infrastructure without having to control or even manage the underlying cloud infrastructure.

## PaaS

**Platform as a service**

The consumer can deploy acquired applications created using programming languages, services, libraries and tools supported by the supplier. They have control over the deployed applications and sometimes the configuration settings for the application and hosting environment.

## IaaS

**Infrastructure as a service**

The consumer can get processing, storage and/or any other computing resources without having to control the underlying infrastructure.

*The distinguishing question in every case: how much of the stack does the consumer control?*

# Who Controls What: The Responsibility Ladder

| Layer | SaaS | PaaS | IaaS |
|---|---|---|---|
| Applications | Provider | Consumer | Consumer |
| Data and configuration | Provider | Shared | Consumer |
| Runtime and middleware | Provider | Provider | Consumer |
| Operating system | Provider | Provider | Shared |
| Servers, storage, network | Provider | Provider | Provider |

*A teaching aid, not an ITIL figure — but it makes the examinable definitions easier to hold in memory.*

# Cloud Service Deployment Models

Every service model can be deployed in several ways — either independently, or using a mix of the following.

1

## Private

Exclusive to one organisation

2

## Public

Open use, provider premises

3

## Community

Shared by related stakeholders

4

## Hybrid

Two or more, bound together

### The critical qualifier: ‘either independently or using a mix’

Service models and deployment models are independent dimensions. An organisation can consume SaaS from a public cloud, run PaaS in a community cloud, and host IaaS privately — all at the same time.

# Private Cloud and Public Cloud

## Private cloud

May be located within the organisation's premises or outside of it. It is a cloud infrastructure or platform to be used exclusively by a specific organisation which, at the same time, can have one or several consumers.

### Ownership and management

Normally managed and owned by an organisation, a provider, or a combination of both. Note that 'private' does not mean 'on premises'.

## Public cloud

Located on the cloud provider premises. It is provisioned for open use and may be owned, managed and operated by any type of organisation interested in using it.

### The defining contrast

Exclusivity of use, not physical location, separates private from public. A hosted private cloud sits on someone else's premises but remains exclusive to one organisation.

# Community Cloud and Hybrid Cloud

## Community cloud

May be owned, managed and operated by one or more of the stakeholders in the community, and may exist on or off the organisation's premises.

### What binds the community

It consists of several cloud services meant to support and share a collection of cloud service customers with the same requirements, who have a relationship with one another. Shared requirements plus an existing relationship — both are needed.

## Hybrid cloud

A composition of two or more distinct cloud infrastructures — private, community or public — that remain unique entities, but are bound together by standardized or proprietary technology.

### What the binding achieves

Data and application portability. The constituent clouds do not merge — they remain unique entities — so hybrid is an integration architecture, not a fifth kind of cloud.

# Deployment Models at a Glance

| Model | Who may use it | Where it sits | Ownership / management | Distinguishing feature |
|---|---|---|---|---|
| **Private** | One specific organisation, which may have one or several consumers | Within the organisation's premises or outside of it | An organisation, a provider, or a combination of both | Exclusivity of use |
| **Public** | Open use — any interested party | On the cloud provider's premises | Any type of organisation interested in using it | Provisioned for open use |
| **Community** | A collection of customers with the same requirements and a relationship with one another | On or off the organisation's premises | One or more of the stakeholders in the community | Shared requirements plus existing relationships |
| **Hybrid** | Depends on the constituent clouds | Across two or more distinct infrastructures | Inherited from each constituent cloud | Bound together for data and application portability |

*Foundation-level examination questions usually turn on exclusivity (private), open use (public), shared requirements (community) and portability (hybrid).*

# The Cloud: Challenge and Opportunity

**The advent of the cloud has been one of the greatest challenges and opportunities within the IT world for decades.**

The promise of rapid, elastic storage and IT services available at the touch of a button is one that many organisations struggle to deliver internally — not because the benefits are not there to be had, but because their own ITSM processes and controls have not been adapted to support a radically different way of working.

### The diagnosis

The obstacle is organisational, not technological. Cloud benefits are real; unadapted processes are what prevent their realisation.

### The reassurance

The management and control of IT services is a key skill of IT departments no matter where those services are physically located.

**The processes and controls offered by ITIL are readily adaptable to support the management of cloud services — adaptation is required, replacement is not.**

# A Joined-Up Response Across Three Levels

A coordinated response to the management of cloud services is essential.

| | |
|---|---|
| **Strategic** | Direction, investment posture and the organisation's own cloud strategy |
| **Tactical** | Supplier arrangements, budgets, capacity thresholds and control design |
| **Operational** | Day-to-day provisioning, incidents, patching and consumption |

**Organisations that treat cloud provision as only an operational issue will suffer on the tactical front; those that control it only tactically will suffer at the strategic level.**

# Six Other Practices Reshaped by Cloud

Apart from infrastructure and platform management, the operation and management of cloud-based services involves many other practices. This is not a comprehensive list.

1. **Service financial management**
2. **Supplier management**
3. **Capacity and performance management**
4. **Change control**
5. **Incident management**
6. **Deployment management**

*Each is examined in turn on the next three slides.*

CLOUD IMPACT 1

# Service Financial Management: CAPEX to OPEX

## CAPEX

Capital expenditure — the traditional model of buying assets outright and depreciating them over their useful life.

## OPEX

Operational expenditure — preferred with cloud, since cloud services are often consumed as utilities and paid out of the operational budget.

### Easy access drives growth

If cloud services are quicker and easier to access than in-house services, associated costs will grow as more parts of the business use them.

### The cost model must change

The IT cost model must be adjusted; fiscal planning that assumed CAPEX will not hold.

### Guard against running dry

Service financial management determines the techniques and controls required so the organisation does not run out of OPEX unexpectedly.

CLOUD IMPACT 2

# Supplier Management and Capacity & Performance

## Supplier management

The focus shifts from simply selecting and onboarding suppliers, to acting as the front end for a full release management process.

*So that these are routinely assessed before onboarding a new cloud offering:*

- **IT security**
- **Data protection**
- **Regulatory compliance**

## Capacity and performance management

Coupled with service financial management, this practice should establish and monitor budgets, with thresholds tracked and warnings published if an upswing in demand leads to an increase in the cost of cloud services.

### The change in kind

On-premises, exceeding capacity degrades performance. In the cloud, it silently increases cost — so capacity management becomes a financial control as well as a technical one.

CLOUD IMPACT 3

# Change Control, Incident and Deployment Management

## Change control

Boundaries must be redefined: cloud service providers often make changes with minimal customer involvement and almost no customer approval. Products and services built on cloud will need far greater use of standard changes to unlock the benefits cloud platforms and their business models provide.

## Incident management

The focus shifts from knowing how to fix in-house issues, to knowing which service is supported by which cloud provider, and what information they will require to resolve an issue. Greater care will be needed to support impacted customers and teams.

## Deployment management

The practice remains critical, but the ability to safely onboard or offboard a cloud provider becomes a common requirement. Deployment management is a key capability for ensuring new cloud capabilities are rapidly deployed and embedded within in-house service offerings.

*Common thread: control shifts from doing the work to governing work done by others.*

FIGURE 5.38

# Infrastructure and Platform Management in the Value Chain

The practice is involved in all value chain activities except engage.




**Contribution scale:** 0 – None | 1 – Low | 2 – Medium | 3 – High

### Plan

Provides information about technology opportunities and constraints used for strategic and tactical planning.

### Improve

Supplies information about technology opportunities supporting continual improvement, and constraints of technologies in use.

### Design and transition

Product and service design benefits from the information provided about technology opportunities and constraints.

### Obtain/build

A critical contributor — it provides necessary information about the components to be obtained.

### Deliver and support

Supports ongoing maintenance of services and infrastructure, including patch management, backups and similar executions.

APPLICATION

# Infrastructure and Platform Decisions in Ghana

### Data residency shapes the deployment model

Ghana's Data Protection Act, 2012 (Act 843) governs how personal data is processed and transferred. Where data must remain under close national control, private or community cloud becomes the practical choice rather than the cheapest one.

### A community cloud for shared sectors

Public universities, teaching hospitals or rural and community banks share requirements and have existing relationships — exactly the conditions the ITIL definition of a community cloud describes.

### Power and connectivity are infrastructure

The text includes buildings and facilities used to run IT infrastructure. In practice this means generators, UPS capacity, cooling and redundant fibre routes are within the practice's scope.

### Foreign-currency exposure in OPEX

Cloud services billed in dollars while budgets are set in cedis convert an IT cost into an exchange-rate risk — a live concern for service financial management.

*Discussion prompt: which deployment model does your own institution use, and was that a decision or an accident?*

# 5.3.3

# Software Development and Management

Ensuring applications meet stakeholder needs in functionality, reliability, maintainability, compliance and auditability.

**Purpose**

**Activities**

**The lifecycle**

**Value chain**

# Purpose of the Practice

**KEY MESSAGE**

The purpose of the software development and management practice is to ensure that applications meet internal and external stakeholder needs, in terms of functionality, reliability, maintainability, compliance, and auditability.

**Functionality** | **Reliability** | **Maintainability** | **Compliance** | **Auditability**

*The five stated quality dimensions — note that three of them are not about what the software does*

**Both internal and external stakeholders count — the developers who maintain it and the auditors who examine it are stakeholders too.**

# What the Term ‘Software’ Covers

Anything from a single program, or suite of programs, to larger constructs — an operating system, an operating environment, or a database — on which various smaller application programs, processes or workflows can run.

*The term includes, but is not limited to:*

**Desktop applications**

**Mobile apps**

**Embedded software**

Controlling machines and devices

**Websites**

**Software applications — whether developed in house or by a partner or vendor — are of critical importance in the delivery of customer value in technology-enabled business services.**

# Activities the Practice Encompasses

1. **Solution architecture**
   The overall structural design of the solution

2. **Solution design**
   User interface, CX, service design and related design work

3. **Software development**
   Building the software itself

4. **Software testing**
   Unit, integration, regression, information security and user acceptance testing

5. **Code repository management**
   Managing repositories or libraries to maintain the integrity of artefacts

6. **Package creation**
   For the effective and efficient deployment of the application

7. **Version control**
   Sharing and ongoing management of smaller blocks of code

# Two Generally Accepted Approaches

## Waterfall

Sequential. Requirements are established up front and each phase completes before the next begins.

- Predictable scope and cost
- Heavy documentation trail
- Change is expensive once underway
- Fits stable, well-understood requirements

## Agile

Iterative and incremental. Working software is produced in short cycles and requirements evolve with feedback.

- Responsive to changing requirements
- Frequent working increments
- Demands close customer involvement
- Fits uncertain or evolving needs

**See section 5.1.8 for fuller treatment of these methods. Neither is presented as universally superior.**

FIGURE 5.39

# The Software Lifecycle

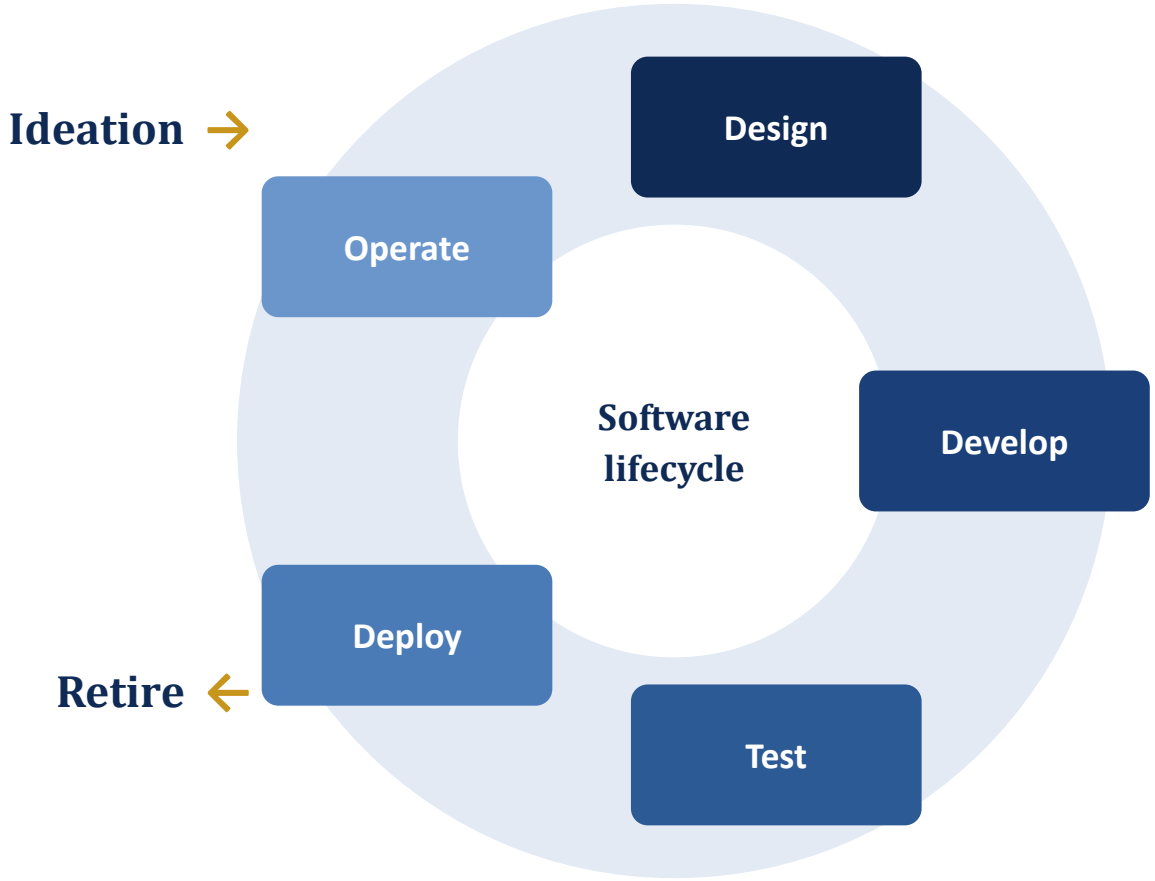
Software components can be continually evaluated using a lifecycle that tracks the component from ideation through to ongoing improvement, and eventually retirement.

**Entry**

An idea enters the loop at design

**Exit**

A component leaves only when retired




**Software management is the wider practice**

It encompasses the ongoing activities of designing, testing, operating and improving software applications so they continue to facilitate value creation — not just the act of building them once.

FIGURE 5.40

# Software Development and Management in the Value Chain

The practice is involved in all value chain activities except engage, with the strongest contribution to obtain/build.




**Contribution scale:** 0 – None; 1 – Low; 2 – Medium; 3 – High

### Plan

Provides information about opportunities and constraints related to the creation and changing of the organisation's software.

### Improve

Service improvements involving software components — especially those developed in house — rely on this practice.

### Design and transition

Allows the organisation to holistically design and manage changes to products and services.

### Obtain/build

The creation of in-house products and the configuration of products developed by partners and suppliers depend on this practice.

### Deliver and support

Provides delivery and support teams with the documentation needed to use products that facilitate the co-creation of value.

SYNTHESIS

# Why None of the Three Contributes to ‘Engage’

All three technical management practices are absent from the engage activity in their heat maps. That is a deliberate statement about the division of labour in the service value chain.

1

### Engage is about stakeholders

It provides a good understanding of stakeholder needs, transparency, and continual engagement with all stakeholders.

2

### Technical practices face inward

They act on components, environments and code — the technology, not the relationship.

3

### Communication is delegated

Release management owns communication about deployments; service level management and the service desk own the stakeholder conversation.

*A useful revision heuristic: if a question asks which value chain activity a technical management practice does not support, engage is the answer.*

SYNTHESIS

# How the Three Practices Work Together

### Software development and management

Builds and packages the component; creates the artefact and the documentation

→

### Infrastructure and platform management

Provides and oversees the environments and platforms the component will run on

→

### Deployment management

Moves the component into those environments — test, staging, then live

**And around them: release management decides when users see it, change control authorises it, and service configuration management records what is where.**

No practice creates value alone. ITIL 4 replaces the process-chain thinking of earlier editions with practices that are combined into value streams — the combination is designed for the situation, not fixed in advance.

# Beyond the text

## Contemporary Issues

The ITIL 4 Foundation text was published in 2019. These developments sit outside the examinable syllabus but shape how the three practices are executed today.

**Platform engineering**

**AI in the toolchain**

**Supply chain security**

**FinOps & sustainability**

CONTEMPORARY ISSUE 1

# From Continuous Delivery to Platform Engineering

The text already anticipates DevOps environments using a complete automated toolchain. The practice has since gone further: many organisations now build an internal developer platform, treating the deployment pipeline itself as a product with its own users.

### GitOps and declarative deployment

Desired state is declared in a repository and reconciled automatically — an extension of infrastructure as code to the whole environment.

### Progressive delivery

Canary releases and feature flags let a change be deployed to everyone but released to a few — sharpening ITIL's deployment/release distinction.

### Golden paths

A curated, paved route to production that makes the compliant option also the easiest one.

### Implication for ITIL

Deployment management increasingly governs a pipeline rather than performing an event — control moves upstream into how the pipeline is designed.

CONTEMPORARY ISSUE 2

# Generative AI in the Software Practice

## Code generation and assistance

AI assistants now draft, complete and refactor code routinely. The practice's stated concerns — maintainability, compliance and auditability — become sharper, not softer, when a human did not write the code.

## Testing and review

AI is used to generate test cases and flag defects. It supplements the testing activities the text lists; it does not replace user acceptance testing, which is a stakeholder judgement.

## AIOps in infrastructure

Anomaly detection and predictive capacity models change what 'monitoring the technology solutions available to the organisation' can mean in practice.

### The governance question this raises

If a defect reaches production in AI-generated code, who is accountable, and can the organisation demonstrate the change was reviewed and authorised? Auditability was always in the key message — AI simply makes it the hard part.

CONTEMPORARY ISSUE 3

# Software Supply Chain Security

## Why the DML matters more than ever

The requirement that components be held in secure locations, unmodified before deployment, was written for a world of internal builds. Modern applications assemble hundreds of third-party dependencies — each one a route into the estate.

## Current responses

- Software bills of materials (SBOM) listing every dependency
- Artefact signing and provenance verification
- Dependency scanning built into the pipeline

### Connecting back to the ITIL text

The supplier management adjustment described for cloud — routinely assessing IT security, data protection and regulatory compliance before onboarding — now applies to open-source dependencies as much as to contracted suppliers. Ghana's Cybersecurity Act, 2020 (Act 1038) and sector directives from the Bank of Ghana push the same obligation onto regulated institutions: know what is in your software, and be able to prove it.

CONTEMPORARY ISSUE 4

# FinOps, Sustainability and the Cost of Elasticity

## FinOps

A discipline that operationalises exactly what the text warns about: cloud costs growing as more of the business consumes services, and the risk of running out of OPEX unexpectedly.

- Showback and chargeback to consuming units
- Rightsizing and reserved-capacity commitments
- Cost as a first-class design constraint
- Joint ownership by finance, engineering and the business

## Sustainable IT

Infrastructure and platform management explicitly includes the buildings and facilities used to run IT infrastructure — which is where energy, cooling and equipment lifecycle decisions live.

- Energy intensity of data centres and cooling
- Workload placement and utilisation efficiency
- E-waste and hardware disposal at retirement
- Reporting expectations from funders and regulators

CONTEMPORARY ISSUE 5

# Digital Sovereignty and Data Residency

Where data physically sits has moved from a technical detail to a board-level and regulatory question — which turns the choice of cloud deployment model into a compliance decision.

### Legal

Ghana's Data Protection Act, 2012 (Act 843) sets conditions for processing and for transferring personal data outside the country.

### Regulatory

Sector regulators — notably the Bank of Ghana for financial institutions — impose additional cyber and information security requirements on outsourced and hosted arrangements.

### Institutional

National infrastructure such as the facilities operated by the National Information Technology Agency offers a domestic hosting option for public-sector workloads.

### Architectural

The practical answer is often hybrid: sensitive records held privately or domestically, elastic workloads in public cloud, bound together for portability.

*Verify current legal and regulatory requirements before applying these points to a live project.*

CONTEMPORARY ISSUE 6

# Serverless, Containers and a Moving Boundary

The three cloud service models remain the examinable framework, but the boundary between them has kept moving since 2019.

### Containers and orchestration

A packaging and runtime layer that sits between IaaS and PaaS — the consumer controls more than PaaS allows, less than IaaS demands.

### Serverless functions

The consumer supplies code and pays per execution. There is no server to manage at all, which stretches the definition of ‘platform’.

### Managed data services

Databases, queues and analytics consumed as services blur where the application ends and the platform begins.

## What has not changed

The organising question the ITIL definitions turn on — how much of the underlying infrastructure must the consumer control or manage? — still separates the models cleanly, whatever the marketing name of the service.

APPLICATION

# Reading the Ghanaian Digital Ecosystem Through 5.3

1

### A national payments platform

Interoperable payment infrastructure has to be deployed to many participating institutions at once when a scheme rule changes — a textbook case where dependencies force a big bang cut-over.

2

### University student information systems

Registration peaks create a demand curve that argues for elastic capacity — but capacity and performance management must publish cost thresholds before, not after, the semester opens.

3

### Public-sector service portals

Government digital services depend on infrastructure and platform management for availability, and on deployment management to push changes without disrupting citizens mid-transaction.

4

### Mobile-first client estates

Where staff work from phones on metered data, pull deployment via a controlled repository is not a preference — it is the only workable approach.

*These are analytical illustrations for classroom discussion, not statements about any named organisation's internal practice.*

REVISION

# Key Terms (1 of 2)

| | |
|---|---|
| **Deployment management** | The practice of moving new or changed hardware, software, documentation, processes or any other component to live environments. |
| **Phased deployment** | Deploying to part of the production environment at a time, repeating until complete. |
| **Continuous delivery** | Integrating, testing and deploying components when they are needed, with frequent customer feedback loops. |
| **Big bang deployment** | Deploying new or changed components to all targets at the same time. |
| **Pull deployment** | Making software available in a controlled repository for users to download to client devices when they choose. |
| **Definitive media library (DML)** | The secure location holding definitive copies of software and documentation available for deployment. |
| **Definitive hardware store (DHS)** | The secure location holding hardware components available for deployment. |

REVISION

# Key Terms (2 of 2)

| | |
|---|---|
| **Infrastructure and platform management** | The practice of overseeing the infrastructure and platforms used by an organisation, including external providers' technology. |
| **SaaS** | Software as a service — use of applications running in the cloud infrastructure without controlling or managing it. |
| **PaaS** | Platform as a service — deploying acquired applications built with supplier-supported languages, services, libraries and tools. |
| **IaaS** | Infrastructure as a service — obtaining processing, storage and other computing resources without controlling the underlying infrastructure. |
| **Private / public cloud** | Exclusive use by one organisation / provisioned on provider premises for open use. |
| **Community / hybrid cloud** | Shared by stakeholders with common requirements and a relationship / two or more distinct clouds bound for data and application portability. |
| **Software development and management** | The practice ensuring applications meet stakeholder needs for functionality, reliability, maintainability, compliance and auditability. |

SELF-ASSESSMENT

# Check Your Understanding (1 of 2)

**1** **Which deployment approach delivers new components to all targets simultaneously?**

A. Phased B. Pull C. Big bang D. Continuous delivery

**2** **Which practice owns communication with users and customers about a deployment?**

A. Deployment management B. Release management C. Change control D. Service desk

**3** **Where are definitive copies of software and documentation held before deployment?**

A. DHS B. CMDB C. DML D. Service catalogue

**4** **In which cloud service model does the consumer have control over deployed applications and sometimes the hosting configuration?**

A. SaaS B. PaaS C. IaaS D. All of them

*Answers: 1 C · 2 B · 3 C · 4 B*

SELF-ASSESSMENT

# Check Your Understanding (2 of 2)

**5** **Which cloud deployment model consists of services shared by customers with the same requirements who have a relationship with one another?**

A. Private B. Public C. Community D. Hybrid

**6** **Which value chain activity do all three technical management practices leave unsupported?**

A. Plan B. Engage C. Improve D. Deliver and support

**7** **With the advent of cloud computing, which expenditure model is preferred?**

A. CAPEX B. OPEX C. Neither D. A fixed blend of both

**8** **Which is NOT listed as an activity of software development and management?**

A. Solution architecture B. Version control C. Package creation D. Negotiating service level agreements

*Answers: 5 C · 6 B · 7 B · 8 D*

TUTORIAL

# Class Activity: The Deployment Decision

**The scenario**

A Ghanaian tertiary institution is replacing its student records system. The new version changes the underlying database schema and is incompatible with the existing mobile app used by 20,000 students. Campuses are spread across three regions, bandwidth varies widely, and the regulator has set a fixed date for the new reporting format.

### In groups of four

Recommend a deployment approach — or a combination — and justify it against release size, type and impact.

### Then identify

Which other practices must be engaged: change control, release management, capacity, supplier and incident management.

### Finally decide

Which cloud deployment model the institution should host on, and defend the choice on data protection grounds.

***20 minutes in groups · 10 minutes plenary · Each group presents one slide of justification***

CONSOLIDATION

# Key Takeaways

1

### Deployment moves components; release makes them available

They are closely related but separate practices. Deployment covers hardware, software, documentation and processes — not software alone.

2

### Four approaches, chosen by fit

Phased, continuous delivery, big bang and pull. Most organisations combine them according to release size, type and impact.

3

### Infrastructure and platform management oversees the whole estate

Physical, virtual and cloud — including external providers' technology, and the buildings that house it.

4

### Cloud changes practices, not the need for them

Service financial, supplier, capacity, change, incident and deployment management all adapt. ITIL's controls are adaptable, not obsolete.

5

### Software development and management is judged on five qualities

Functionality, reliability, maintainability, compliance and auditability — across a lifecycle running from ideation to retirement.

SOURCES

# References and Further Reading

### Primary source for this session

AXELOS (2019). ITIL® Foundation: ITIL 4 Edition. London: TSO. Chapter 5, sections 5.3–5.3.3, pp. 160–167, including Figures 5.37, 5.38, 5.39 and 5.40.

### Related ITIL 4 material

AXELOS (2019). ITIL® 4: Create, Deliver and Support; ITIL® 4: High-velocity IT — for fuller treatment of deployment, software development and the DevOps toolchain.

### Cloud definitions

Mell, P. and Grance, T. (2011). The NIST Definition of Cloud Computing. NIST Special Publication 800-145 — the source of the service and deployment model taxonomy ITIL adopts.

### Ghanaian legal and regulatory context

Data Protection Act, 2012 (Act 843); Cybersecurity Act, 2020 (Act 1038); Bank of Ghana Cyber and Information Security Directive.

### Note on contemporary issues

Slides 54–61 extend beyond the 2019 ITIL 4 Foundation text and are not examinable at Foundation level. Verify currency of legal and regulatory references before citing them in assessed work.

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End of session

# Questions and Discussion

Next: review sections 5.3–5.3.3 in the ITIL 4 Foundation text, then attempt the eight self-assessment questions without referring to your notes.

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